



UNIVERSITI TEKNOLOGI MALAYSIA

SECR1033 - COMPUTER ORGANIZATION AND ARCHITECTURE

PROJECT 2

NO	NAME	MATRIC NO
1.	MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI	A23CS250
2.	IMAN ABADI BIN MOHD NIZWAN	A23CS0084
3.	MOHAMED ALIF FATHI BIN ABDUL LATIF	A23CS112

Video Link benchmark tools demo : https://youtu.be/xoCB_xPWhDQ

Video Link assembly code demo : <https://youtu.be/TKvjVO9BFbo>

Video Link presentation : <https://youtu.be/D098XOHQrSs>

TABLE OF CONTENT

1.0 INTRODUCTION

1.1 About CPU Benchmarking

1.1 Problem Background

1.2 Aim

1.3 Objective

1.4 Scopes (computer specification & benchmarking tools features)

2.0 FLOWCHART OF THE PROJECT

3.0 BENCHMARKING RESULT

4.0 ANALYSIS, COMPARISON AND DISCUSSION

5.0 CONCLUSION AND REFLECTION

5.1 Conclusion

5.2 Reflection

5.3 References

5.3 Appendices

1.0 INTRODUCTION

About CPU Benchmarking

A benchmark for central processing units (CPUs) is a collection of tests aimed to assess and compare the capabilities of different CPUs. These tests use a variety of computing tasks and processes to assess processing speed, efficiency, and capability. Users can evaluate and compare various processors using CPU benchmark results as a standard reference to help them choose the hardware that best suits their needs.

There are three types of CPU benchmarking methods which are synthetic benchmarking, real-world benchmarking and application benchmarking. When attempting to quickly and thoroughly compare different CPUs, synthetic benchmarks are used. A wide range of operations, including web surfing, file compression, 3D rendering, floating-point calculations, and more, are simulated using synthetic tests. The CPU performance levels at each task are measured, and the results are weighted before they are combined into a single score. Because the tests used to determine the scores are simulations, the results are considered "synthetic." They replicate the workload an application may apply on a CPU under various conditions, as opposed to testing the processor's performance in a particular program (such a game or 3D creation suite). Next, real-world benchmarking is used when specific tasks are needed to run on a PC and require a precise performance indicator for specific programs. In order to run these tests, real programs are given challenging tasks, and the completion times are recorded. Because of this, they offer a reliable first look at the performance under similar conditions. Application benchmarks are actual, fully functional applications that are frequently used to address a range of computational problems, although they can be difficult to establish and port. The most often used benchmark for evaluating CPU performance is SPEC CPU.

Problem Background

Benchmarking is a relatively new concept among computer science students. Students that have no prior knowledge of benchmarking might see them as random numbers when running a program. However those numbers provide significant meaning on what there is to know about the CPU in a computer. This can especially be helpful for students when they want to buy next computers that suit their requirement to run specific programs. For example, using a low performance CPU can hinder students when they want to run programs with a high CPU workload. Thus, the knowledge on how to execute CPU benchmarks and compare them can significantly help them when trying to buy a new computer that fulfils their needs.

Aim

The aim of this project is to learn on CPU benchmarking using benchmarking softwares to conduct a performance benchmarking on multiple computers to analyse and compare the result. Other than that, write an assembly language program to calculate and measure the time taken to execute different loops using a polynomial equation as a benchmarking algorithm.

Objective

- Understand the principles of CPU benchmarking
- Compare the performance of different CPUs
- Analyse the impact of different workloads on CPU performance
- Learn to use free benchmarking tools and interpret their results
- Write and use the assembly code program as benchmarking software

Scopes (computer specification & benchmarking tools features)

For computers to be benchmarked, we use Asus TUF F15, Lenovo Legion Y7000 and HP Pavilion 15

Specification	Computer type		
	Asus TUF F15	Lenovo Legion Y7000	HP Pavilion 15
Processor type	Intel Core i5 11400H	Intel Core i5 9300H	Intel Core i7 1065G7
RAM size	16 GBytes	8 GBytes	16 GBytes
GPU type	Nvidia GeForce RTX 2050	NVIDIA GeForce GTX 1650	NVIDIA GeForce MX250
Motherboard	ASUS FX506HF	LENOVO LNVNB161216	HP 86E2
Bus specs	PCI-Express 4.0 (16.0 GT/s)	PCI-Express 3.0 (8.0 GT/s)	PCI-Express 3.0 (8.0 GT/s)

In this project, we use CPU-Z, Geekbench, Windows Task Manager, MiniTool Partition Wizard and Assembly Language code as our benchmarking tool.

1. CPU-Z

CPU-Z is a popular utility software that provides detailed information about the hardware components of a computer system. It is primarily used to gather information about the CPU (Central Processing Unit), but it also provides information about other system components such as:

- Processor name and number, codename, process, package, cache levels.
- Mainboard and chipset.
- Memory type, size, timings, and module specifications (SPD).
- Real time measurement of each core's internal frequency, memory frequency.
- CPU-Z is fully supported on Windows® 11

2. Geekbench

Geekbench is a cross-platform benchmarking tool that provides detailed performance metrics for computer systems. Here is a summary of the main uses of Geekbench:

The program handles:

- Add system info as complement /conformation to CPU-Z o
- CPU test with variation of technique or method used, either single core or multi-core performance.

3. Windows Task Manager

Windows Task Manager is a system monitoring tool that provides real-time information about the computer's performance and running applications. Here is a summary of the main uses of Windows Task Manager.

The program handles:

- Process Management: Displays all running processes and their resource usage.
- Performance Monitoring: Provides real-time statistics on CPU, memory, disk, and network usage.
- Application Management: Allows you to end tasks or switch to running applications.
- Startup Management: Manages which programs run on startup.
- Service Management: Displays and manages system services.
- User Management: Shows users currently logged into the system and their resource usage.

- Resource Monitoring: Detailed views of resource consumption by individual processes.

4. MiniTool Partition Wizard

MiniTool Partition Wizard is a disk partition management tool that allows users to manage disk partitions with various features. Here is a summary of the main uses of MiniTool Partition Wizard:

The program handles:

- Disk Partitioning: Creates, deletes, formats, and resizes partitions.
- Partition Recovery: Recovers lost or deleted partitions.
- Disk Cloning: Clones entire disks or specific partitions to another disk.
- Disk Checking: Checks and fixes file system errors.
- Disk Benchmarking: Measures the performance of your disks.
- Data Backup: Provides tools for backing up important data.

5. Assembly Language

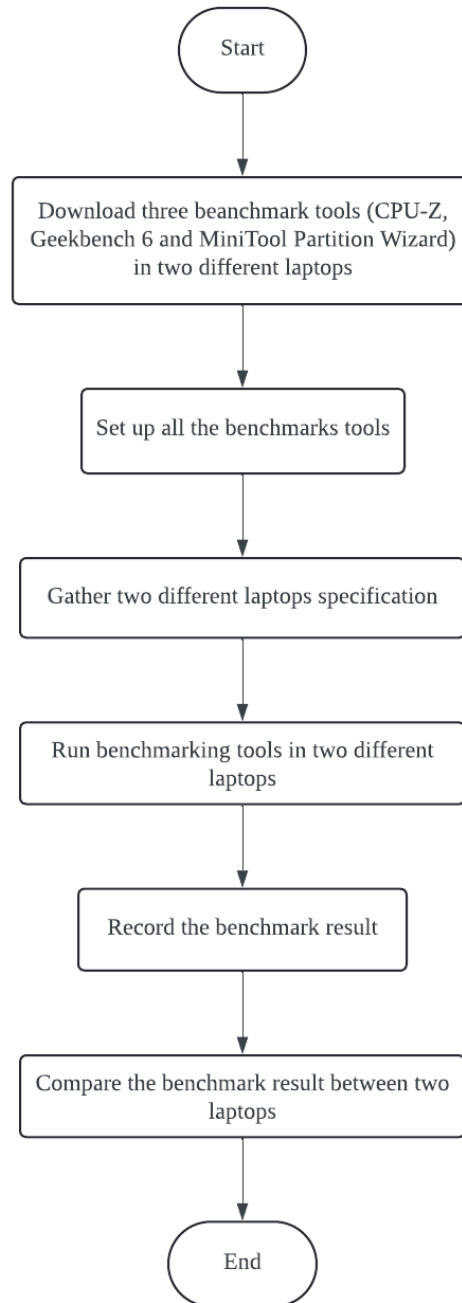
An assembly language program was written to calculate and measure the time taken to execute different loops for the polynomial equation:

$$y = (a * x^3) + (b * x^2) + c*x + d$$

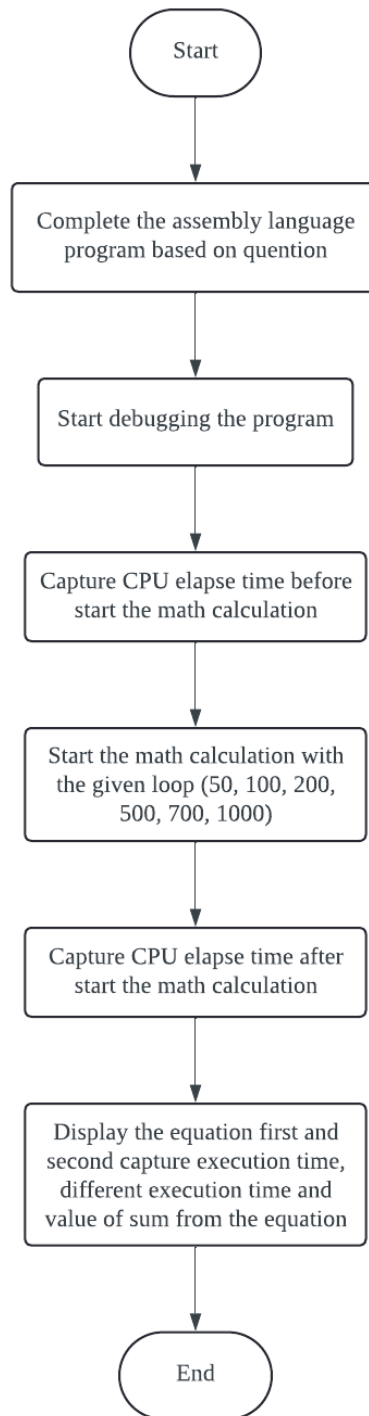
where a, b, c, d are coefficients of delay and x is an integer input value. Implement a loop that iterates different loop times, each time calculating the value of y for a different x value. Different loops will represent different loads that will be imposed on the polynomial equation as a benchmarking method. Measurements of the time taken to complete the different loop and output was recorded for comparison.

2.0 FLOWCHART

PART 1 :



PART 2 :



3.0 BENCHMARKING RESULT

PART 1 :

CPU-Z

Benchmark		Computer Type		
		Asus TUF F15	Lenovo Legion Y7000	HP Pavilion 15
CPU processor		Intel Core i5 11400H	Intel Core i5 9300H	Intel Core i7 1065G7
Clock speed		4089.36 MHz	2395.02 MHz	1297.14 MHz
CPU Technology		10 nm	14 nm	10 nm
Cache	L1 Data	48 KB x 6 12-way	32 KB x 4 8-way	48 KB x 4 12-way
	L1 Instruction	32 KB x 6 8-way	32 KB x 4 8-way	32 KB x 4 8-way
	L2	1.25 MB x 6 20-way	256 KB x 4 4-way	512 KB x 4 8-way
	L3	12 MB x 1 8-way	8 MB x 1 16-way	8 MB x 1 16-way
Memory	Type	DDR4	DDR4	DDR4
	Size	16 GB	8 GB	16 GB
	Channel	Dual	Single	Dual
Motherboard	Model	FX506HF	LNVNB161216	86E2
	Bus Specs	PCI-Express 4.0 (16GT/s)	PCI-Express 3.0 (8GT/s)	PCI-Express 3.0 (8GT/s)

GeekBench

Benchmark		Computer Type		
		Asus TUF F15	Lenovo Legion Y7000	HP Pavilion 15
CPU Processor		Intel Core i5 11400H	Intel Core i5 9300H	Intel Core i7 1065G7
Base Clock Speed		2.69 GHz	2.40 GHz	1.50 GHz
Maximum Clock Frequency		4491 MHz	4097 MHz	3891 MHz
Cache	L1 Data	48.0 KB x 6	32.0 KB x 4	48.0 KB x 4
	L1 Instruction	32.0 KB x 6	32.0 KB x 4	32.0 KB x 4
	L2	1.25 MB x 6	256 KB x 4	512 KB x 4
	L3	12.0 MB x 1	8.00 MB x 1	8.00 MB x 1
Memory	Type	DDR4 SDRAM	DDR4 SDRAM	DDR4 SDRAM
	Size	16 GB	8 GB	16 GB
	Channel	2	1	2
	Transfer Rate	3194 MT/s	2660 MT/s	2660 MT/s
Motherboard Model		FX506HF	LNVNB161216	86E2
CPU Test (Single-Core Performance)	File Compression	292.4 MB/s	191.2 MB/s	238.0 MB/s
	Navigation	10.8 routes/s	10.1 routes/s	10.2 routes/s
	HTML5 Browser	43.7 pages/s	30.6 pages/s	37.8 pages/s
	PDF Renderer	49.7 Mpixels/s	35.8 Mpixels/s	40.7 Mpixels/s
	Photo Library	28.1 images/s	16.1 images/s	23.3 images/s

Task Manager

Benchmark		Computer Type		
		Asus TUF F15	Lenovo Legion Y7000	HP Pavilion 15
Application Opened	Low Load	- Google Chrome (1 tab) - Microsoft Word	Microsoft Word	- Google Chrome (1 tab) - Microsoft Word
	Medium Load	- Google Chrome (4 tab) - Microsoft Word - Microsoft Excel - Microsoft Powerpoint - Spotify - Adobe Acrobat	- Google Chrome (3 tab) - Microsoft Word - Microsoft Excel	- Google Chrome (6 tab) - Microsoft Word - Microsoft Excel - Microsoft Powerpoint - Telegram Desktop - Github Desktop - VS Code - Discord
	High Load	- Google Chrome (4 tab) - Microsoft Word - Microsoft Excel - Microsoft Powerpoint - Spotify - Adobe Acrobat - Steam - Goat Simulator	- Google Chrome (3 tab) - Microsoft Word - Microsoft Excel - League of Legend	- Google Chrome (8 tab) - Microsoft Word - Microsoft Excel - Microsoft Powerpoint - Telegram Desktop - Github Desktop - VS Code - Discord - Webex Desktop - Adobe Acrobat Reader - RStudio - Call of Duty Modern warfare 3 2011

MiniTool Partition Wizard

Test Mode		Transfer Size	Computer Type		
			Asus TUF F15	Lenovo Legion Y7000	HP Pavilion 15
Sequential	Read	4 KB	1087.13 MB/s	736.17 MB/s	46.58 MB/s
		8 KB	1241.41 MB/s	530.16 MB/s	66.12 MB/s
		12 KB	1138.61 MB/s	758.69 MB/s	127.64 MB/s
		32 KB	1131.22 MB/s	1303.22 MB/s	181.88 MB/s
	Write	4 KB	937.01 MB/s	664.35 MB/s	306.85 MB/s
		8 KB	1400.03 MB/s	1141.82 MB/s	384.85 MB/s
		12 KB	1427.08 MB/s	1391.18 MB/s	401.03 MB/s
		32 KB	1570.83 MB/s	832.16 MB/s	439.22 MB/s
Random	Read	4 KB	1006.94 MB/s	844.10 MB/s	117.11 MB/s
		8 KB	1307.17 MB/s	502.63 MB/s	46.85 MB/s
		12 KB	1421.35 MB/s	939.72 MB/s	40.64 MB/s
		32 KB	1161.49 MB/s	1428.69 MB/s	112.12 MB/s
	Write	4 KB	905.34 MB/s	735.99 MB/s	95.04 MB/s
		8 KB	1111.87 MB/s	1130.58 MB/s	1.293 MB/s
		12 KB	1124.94 MB/s	1390.26 MB/s	372.49 MB/s
		32 KB	1276.09 MB/s	1472.98 MB/s	423.78 MB/s

PART 2 :

In this part, an assembly code was used to do the benchmarking process of the CPU capture time. The result of the benchmark is based off of this assembly code written below:

TITLE Project COA

; Author: Iman Abadi Bin Mohd Nizwan (Birthday: 10/5/2004)

; : Mohamed Alif Fathi Bin Abdul Latif (Birthday: 31/12/2004)

; : Muhammad Mukhritz Al-Iman Bin Raffi (Birthday: 16/1/2004)

include Irvine32.inc

.data

message1 BYTE "Welcome to CPU Benchmark Program",0dh,0ah,0dh,0ah

BYTE "Benchmark CPU time Using Equation $y = (10 * x^3) + (31 * x^2) + (16 * x) + 1$ ",
0dh,0ah

BYTE " (with only coef1, coef2, coef3, coef4 = 10, 31, 16, 1 msec)", 0dh,0ah,0dh,0ah,0

message2 BYTE "Enter the Number of Looping (N) = ", 0

message3 BYTE "CPU time Stress Test in progress...",0dh,0ah,0dh,0ah,0

message4 BYTE "Result:",0dh,0ah,0dh,0ah

BYTE "First Capture Execution time in milisecond: ", 0

message5 BYTE "Second Capture Execution time in millisecond: ", 0

message6 BYTE "Different Execution time in millisecond: ", 0

message7 BYTE "Value of Sum from the Stress Test (polynomial) = ", 0

message8 BYTE "Press 'y' to continue or 'n' to exit the benchmark : ", 0

max_loop DWORD ?

msec1 DWORD ?

msec2 DWORD ?

sum DWORD 0

coef1 DWORD 10

coef2 DWORD 31

coef3 DWORD 16

coef4 DWORD 1

initial_value DWORD 1

stryn BYTE "Press 'y' to Main Menu or 'n' to Exit Program : ", 0

charIn BYTE ?

charY db 'y'

strbye BYTE "Thank you ... BYE!!", 0dh, 0ah, 0

.code

main proc

startProg:

call Clrscr

mov edx,OFFSET message1

call WriteString

mov edx,OFFSET message2

call WriteString

call ReadDec

mov max_loop, eax

mov edx, OFFSET message3

call WriteString

call GetMseconds

mov msec1, eax

mov ecx, max_loop

L1:

; (10 * x^3)

mov sum, 0d

```
mov eax, initial_value
mov ebx, initial_value
mul ebx ; eax = eax * ebx
mul ebx ; eax = eax * ebx
mov ebx , 10d
mul ebx ; eax = eax * ebx
add sum , eax
```

```
; (31 * x^2)
```

```
mov eax, initial_value
mov ebx, initial_value
mul ebx ; eax = eax * ebx
mov ebx , 31d
mul ebx ; eax = eax * ebx
add sum , eax
```

```
; (16 * x)
```

```
mov eax , initial_value
mov ebx , 16d
mul ebx ; eax = eax * ebx
add sum , eax
```

```
; + 1
```

```
add sum , 1d
inc initial_value
call Delay
loop L1
```

```
call GetMseconds
mov msec2, eax
```

```
mov edx, OFFSET message4
call WriteString
mov eax , msec1
call WriteDec
call crlf
```

```
mov edx, OFFSET message5
call WriteString
mov eax , msec2
call WriteDec
call crlf
```

```
mov edx, OFFSET message6
call WriteString
mov eax , msec2
sub eax , msec1
call WriteDec
call crlf
```

```
mov edx, OFFSET message7
call WriteString
mov eax , sum
call WriteDec
call crlf
call crlf
```

```
mov edx, OFFSET message8
call WriteString
call ReadChar
mov charIn, AL
call WriteChar
call Crlf
call Crlf
mov BL, charY
cmp BL, charIn
JE startProg
```

```
mov edx, OFFSET strbye
call WriteString
```

```
exit
main ENDP
END main
```

The CPU elapsed time was captured using the *call GetMsecods* that returns the number of milliseconds that have elapsed since midnight. The capture time was done before the execution of the mathematical equation and then after the execution. From there the difference of the capture time can be calculated. Then, the steps are repeated for a different number of loops.

Results of the Benchmark

50 Loops

```
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = (10 * x^3) + (31 * x^2) + (16 * x) + 1$ 
      (with only coef1, coef2, coef3, coef4 = 10, 31, 16, 1 msec)

Enter the Number of Looping (N) = 50
CPU time Stress Test in progress...

Result:

First Capture Execution time in milisecond: 83706958
Second Capture Execution time in millisecond: 83727790
Different Execution time in millisecond: 20832
Value of Sum from the Stress Test (polynomial) = 1328301

Press 'y' to continue or 'n' to exit the benchmark :
```

100 Loops

```
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = (10 * x^3) + (31 * x^2) + (16 * x) + 1$ 
      (with only coef1, coef2, coef3, coef4 = 10, 31, 16, 1 msec)

Enter the Number of Looping (N) = 100
CPU time Stress Test in progress...

Result:

First Capture Execution time in milisecond: 83809287
Second Capture Execution time in millisecond: 83890994
Different Execution time in millisecond: 81707
Value of Sum from the Stress Test (polynomial) = 10311601

Press 'y' to continue or 'n' to exit the benchmark : n
```

200 Loops

```
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = (10 * x^3) + (31 * x^2) + (16 * x) + 1$ 
      (with only coef1, coef2, coef3, coef4 = 10, 31, 16, 1 msec)

Enter the Number of Looping (N) = 200
CPU time Stress Test in progress...

Result:

First Capture Execution time in milisecond: 83930306
Second Capture Execution time in millisecond: 84253496
Different Execution time in millisecond: 323190
Value of Sum from the Stress Test (polynomial) = 81243201

Press 'y' to continue or 'n' to exit the benchmark :
```

500 Loops

```
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = (10 * x^3) + (31 * x^2) + (16 * x) + 1$ 
      (with only coef1, coef2, coef3, coef4 = 10, 31, 16, 1 msec)

Enter the Number of Looping (N) = 500
CPU time Stress Test in progress...

Result:

First Capture Execution time in milisecond: 84285256
Second Capture Execution time in millisecond: 86293360
Different Execution time in millisecond: 2008104
Value of Sum from the Stress Test (polynomial) = 1257758001

Press 'y' to continue or 'n' to exit the benchmark :
```

700 Loops

```
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = (10 * x^3) + (31 * x^2) + (16 * x) + 1$ 
      (with only coef1, coef2, coef3, coef4 = 10, 31, 16, 1 msec)

Enter the Number of Looping (N) = 700
CPU time Stress Test in progress...

Result:

First Capture Execution time in milisecond: 346326
Second Capture Execution time in millisecond: 5112930
Different Execution time in millisecond: 4766604
Value of Sum from the Stress Test (polynomial) = 3445201201

Press 'y' to continue or 'n' to exit the benchmark :
```

1000 Loops

```
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = (10 * x^3) + (31 * x^2) + (16 * x) + 1$ 
      (with only coef1, coef2, coef3, coef4 = 10, 31, 16, 1 msec)

Enter the Number of Looping (N) = 1000
CPU time Stress Test in progress...

Result:

First Capture Execution time in milisecond: 40070404
Second Capture Execution time in millisecond: 48093723
Different Execution time in millisecond: 8023319
Value of Sum from the Stress Test (polynomial) = 1441081409

Press 'y' to continue or 'n' to exit the benchmark :
```

4.0 ANALYSIS, COMPARISON AND DISCUSSION

PART 1 :

After running the benchmark test on three CPUs, we get the required performance data. Using obtained data, we can obtain detailed information about CPU performance.

CPU-Z

1. Clock speed

The clock speeds of different computer models can significantly impact their performance, as demonstrated by the following examples: the Asus TUF F15, Lenovo Legion Y7000, and HP Pavilion 15. The Asus TUF F15 stands out with a clock speed of 4,089.36 MHz, indicating a robust capacity for handling high-performance tasks and gaming applications. In contrast, the Lenovo Legion Y7000 operates at a clock speed of 2,395.02 MHz, which, while lower than the Asus TUF F15, still provides strong performance for general computing and mid-range gaming. The HP Pavilion 15, with a clock speed of 1,297.14 MHz, represents a more modest option, suitable for everyday tasks such as web browsing, word processing, and media consumption. This comparison highlights the differences in clock speeds among these models, reflecting their intended use cases and performance capabilities.

2. Memory size (RAM)

After examining the specifications, we found that the ASUS TUF F15 and HP Pavilion 15 both have 16 GB of memory, while the Lenovo Legion Y7000 has 8 GB of memory. Memory size refers to the amount of RAM available for the system to use for running applications and processes. The difference in memory size between the ASUS TUF F15, HP Pavilion 15, and Lenovo Legion Y7000 impacts their ability to handle multiple tasks and run memory-intensive applications smoothly. Generally, more memory allows for better multitasking and faster performance, as the system can store more data for quick access. With 16 GB of memory, the ASUS TUF F15 and HP Pavilion 15 may offer better performance in demanding scenarios compared to the Lenovo Legion Y7000 with 8 GB of memory. However, it is important to consider other factors such as CPU and GPU performance, which also significantly influence the overall performance of a laptop.

3. Cpu technology

After examining the specifications, we found that the ASUS TUF F15 and HP Pavilion 15 both utilize 10nm CPU technology, while the Lenovo Legion Y7000 uses 14nm CPU technology. CPU technology, specifically the manufacturing process node size (measured in nanometers), indicates the size of the transistors used in the processor.

Smaller process nodes, like 10nm, generally allow for more transistors on a chip, leading to potential improvements in performance and energy efficiency. The difference in CPU technology between the ASUS TUF F15, HP Pavilion 15, and Lenovo Legion Y7000 suggests that the ASUS and HP models might have advantages in terms of power efficiency and processing power due to their more advanced 10nm technology. However, overall performance also depends on other factors, such as the CPU architecture and clock speed.

Geekbench

When comparing the clock speed results of three laptops between the CPU-Z, and Geekbench, the Asus TUF F15, Lenovo Legion Y7000, and HP Pavilion 15 using CPU-Z and Geekbench, we see differences in reported values. For the Asus TUF F15, CPU-Z reports 4.09 GHz, while Geekbench shows 2.69 GHz. The Lenovo Legion Y7000 has a consistent 2.40 GHz in both CPU-Z and Geekbench. For the HP Pavilion 15, CPU-Z reports 1.30 GHz, whereas Geekbench reports 1.50 GHz. These differences highlight that CPU-Z measures peak performance, while Geekbench focuses on base operational frequencies.

When comparing the single-core performance of the Asus TUF F15, Lenovo Legion Y7000, and HP Pavilion 15 using Geekbench across multiple tasks, the Asus TUF F15 consistently outperforms the others, reflecting its robust hardware capabilities. In file compression, the Asus TUF F15 achieves 292.4 MB/s, significantly higher than the Lenovo Y7000's 191.2 MB/s and the HP Pavilion 15's 238.0 MB/s, indicating superior processing power for handling compressed data. For navigation tasks, the Asus TUF F15 processes 10.8 routes/s, slightly better than the Lenovo Y7000's 10.1 routes/s and the HP Pavilion 15's 10.2 routes/s, suggesting a marginally faster response in pathfinding algorithms. In HTML5 browser performance, the Asus TUF F15 handles 43.7 pages/s, surpassing the Lenovo Y7000's 30.6 pages/s and the HP Pavilion 15's 37.8 pages/s, demonstrating a quicker rendering of web pages. This data highlights the superior single-core performance of the Asus TUF F15 across various tasks, making it a better choice for users requiring high-performance computing in diverse applications.

Task Manager

For low load tasks, the Asus TUF F15 and HP Pavilion 15 both handle running Google Chrome with one tab and Microsoft Word. In contrast, the Lenovo Legion Y7000 is limited to running only Microsoft Word. This indicates that both the Asus TUF F15 and HP Pavilion 15 have slightly higher capacity for multitasking at a basic level compared to the Lenovo Legion Y7000

When it comes to medium load tasks, the Asus TUF F15 manages 4 Chrome tabs along with Microsoft Word, Excel, and PowerPoint, Spotify, and Adobe Acrobat. The Lenovo Legion Y7000 handles a lighter load with 3 Chrome tabs and the same suite of Microsoft Office applications. The HP Pavilion 15 shows superior performance by managing 6 Chrome tabs, the Microsoft Office suite, Telegram Desktop, Github Desktop, VS Code, and Discord. This makes the HP Pavilion 15 the most versatile for medium load multitasking among the three

Under high load conditions, the Asus TUF F15 supports a wide range of applications including 4 Chrome tabs, the full Microsoft Office suite, Spotify, Adobe Acrobat, Steam, and Goat Simulator. The Lenovo Legion Y7000, while still capable, focuses on running 3 Chrome tabs, Microsoft Word, Excel, and League of Legends. The HP Pavilion 15 outperforms the others by handling 8 Chrome tabs, the complete Microsoft Office suite, multiple communication and development tools, Webex Desktop, Adobe Acrobat Reader, RStudio, and Call of Duty Modern Warfare 3 (2011). This makes the HP Pavilion 15 the most robust option for high load scenarios, demonstrating its superior multitasking and processing capabilities.

MiniTool Partition Wizard

In MiniTool Partition Wizard, we tested two modes which are sequential and random. For each mode, we tested the reading speed and writing speed as below:

Sequential Read Performance

For the sequential read benchmark, Asus TUF F15 scored better than the rest at almost all transfer sizes. Setting 4 KB, it would make it to 1087.13 MB/s while Lenovo Legion Y7000 at 736.17 MB/s and HP Pavilion 15 at 46.58MB/s. This trend continues with 8 KB (1241.41 MB/s for Asus, 530.16 MB/s for Lenovo, 66.12 MB/s for HP), 12 KB (1138.61 MB/s for Asus, 758.69 MB/s for Lenovo, 127.64 MB/s for HP), and 32 KB (1131.22 MB/s for Asus, 1303.22 MB/s for Lenovo, 181.88 MB/s for HP), where the lead in most of the speeds is by the Asus TUF F15, except for the 32 KB one, where Lenovo leads.

Sequential Write Performance

In general, the Asus TUF F15 also manages to come out on top in sequential write tests. It scores 937.01 MB/s for 4 KB writes and beats Lenovo's 664.35 MB/s and HP's 306.85 MB/s. At 8 KB, the Asus notebook attains 1400.03 MB/s, with Lenovo trailing at 1141.82 MB/s and HP running away to 384.85 MB/s. With 12 KB writes, it is the same case, with Asus leading at 1427.08 MB/s, followed by Lenovo at 1391.18 MB/s and HP at. Likewise, at 32 KB, Asus hits 1570.83 MB/s with Lenovo coming in at 832.16 MB/s and HP hitting 439.22 MB/s in this mode.

Random Read Performance

In random read performance, the Asus TUF F15 has faster speeds at smaller transfer sizes compared to the Lenovo but the Lenovo is faster at larger sizes. At 4 KB, the Asus reaches 1006.94 MB/s while the Lenovo is at 844.10 MB/s and HP at 117.11 MB/s. For 8 KB, Asus does 1307.17 MB/s while Lenovo sees a haul of 502.63 MB/s, and HP demonstrates 46.85 MB/s. For 12 KB, Asus gets 1421.35 MB/s to Lenovo's 939.72 MB/s and HP's 40.64 MB/s. For 32 KB, Lenovo is the champion with 1428.69 MB/s, more than Asus's 1161.49 MB/s and HP's 112.12 MB/s.

Random Write Performance

On random write tests, the Asus TUF F15 can still keep a performance lead. For 4 KB writes, Asus is at 905.34 MB/s, Lenovo is at 735.99 MB/s, and HP is at 95.04 MB/s. At 8 KB, it is 1111.87 MB/s for Asus, then Lenovo at 1130.58 MB/s, and HP at 1293

MB/s. With 12 KB, it is 1124.94 MB/s for Asus, 1390.26 MB/s for Lenovo, and 372.49 MB/s for HP. With 32 KB, Asus clocks at 1276.09 MB/s, Lenovo tops at 1472.98 MB.

In summary, Asus TUF F15 consistently outperforms the Lenovo Legion Y7000 and HP Pavilion 15 in most scenarios in testing sequential and random read/write performance using MiniTool Partition Wizard. For sequential reads and writes, Asus leads at almost all transfer sizes, though Lenovo surpasses Asus in some larger transfer sizes. In random reads, Asus excels at smaller sizes, while Lenovo is faster at larger sizes. Random writes show a mixed performance, with Asus and Lenovo trading places depending on the transfer size. Overall, the Asus TUF F15 generally demonstrates superior performance across various transfer sizes and tasks compared to its competitors.

PART 2 :

After running the the assembly code with different max_loop (N) (50, 100, 200, 500, 700, 1000), the results of capture time (before), capture time (before) and difference in capture time is in the table below:

No of Loop (N)	Capture_msec (before)	Capture_msec (before)	CPU_msec (difference)
50	83706958	83727790	20832
100	83809287	83890994	81707
200	83930306	84253496	323190
500	84285256	86293360	2008104
700	346326	5112930	4766604
1000	40070404	48093723	8023319

Table 4.1 : Comparison Table of CPU Capture Time

From **Table 1.4**, the time it took to capture the CPU before and after execution of a certain calculation that involves a certain number of loops (N). Thus, the capture time stated in milliseconds corresponds to each of the specified loop count, for before and after executing the mathematical equation and the two of them are subtracted. This comparison assists in assessing the effect to be caused by a higher loop count on the CPU time.

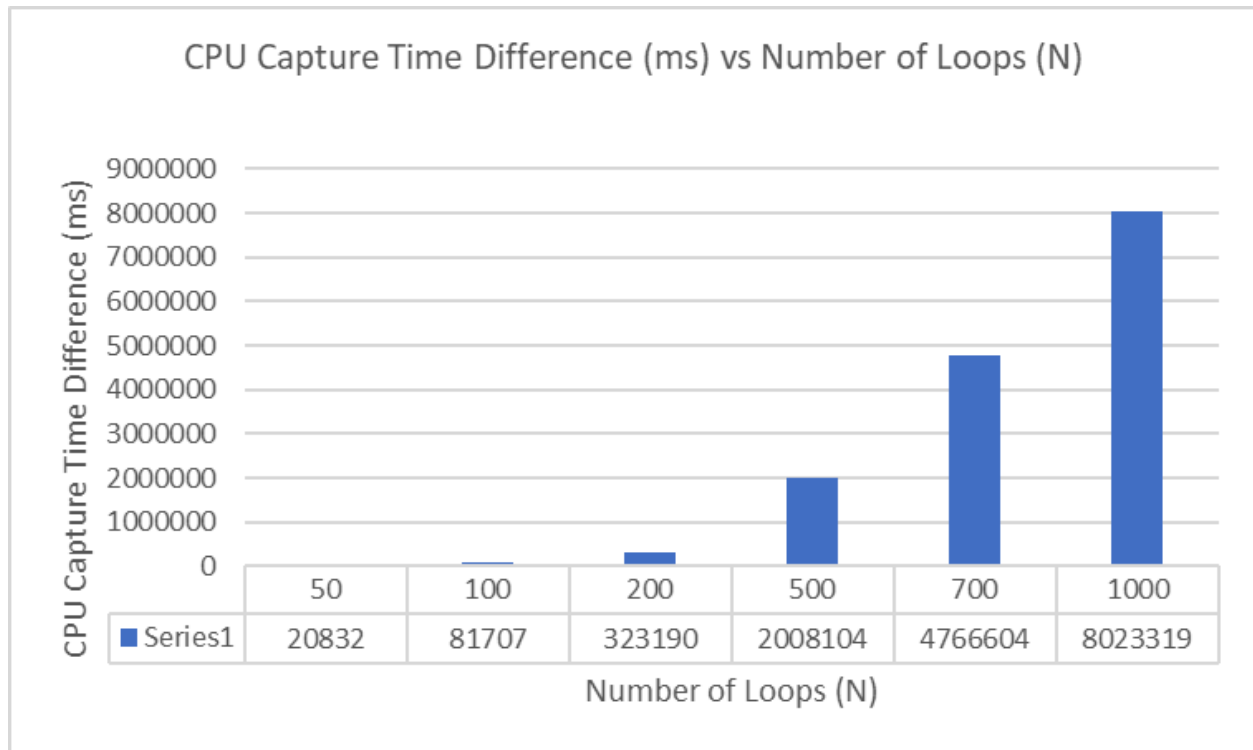


Figure 4.1 : Graph of CPU Capture Time Difference (ms) vs Number of Loops (N)

From **Figure 4.1**, scenarios show that when increasing the number of loops, the capture time before and after the code is also increasing. For instance, with 50 loops, the time difference is at 20832 msec that drastically increases to 8023319 msec 1000 loops. This infers that for higher loop counts the CPU takes longer time to process it, which clearly means the complexity level in terms of computational processes has gone up. The high difference of capture time at the increased number of loops of each loop count, such as 200, 500, 700, and 1000, proves the fact that increasing iterations calls for significantly more CPU time as the result of the operations to be executed is far more complex than the prior loop count number.

5.0 CONCLUSION AND REFLECTION

5.1 Conclusion

In conclusion, to measure the performance of a computer we do the CPU benchmark using multiple tools which are CPU-Z, Geekbench, Window Task Manager and MiniTool Partition Wizard Free 12.8. We use the CPU-Z for collecting spec of our laptop such as our laptop CPU processor, cache, memory and also our motherboard. Next, we use the Geekbench for running CPU stress tests. We test for method file compression, navigation and HTML5 browser for the single core. From that, we obtain the speed for every method that we choose. After that, we use the window task manager to explore the load of running app and computer resources. We test for three categories which are low, middle and high. For low, we open multiple apps until the memory consumption is around 30% - 50%. For middle categories, we open multiple apps until the memory consumption is around 50% to 60%. Lastly, for the high load we open multiple apps until memory consumption is above 70%. For the MiniTool Partition Wizard Free 12.8 is to perform a disk benchmarking process. Finally, from the assembly code, the difference in CPU capture time provided knowledge on how higher workloads affect the CPU capture time

This comparison highlights the important role of hardware specification in affecting the overall performance of the computer. Each computer has their own unique specification that differs from others. As an example, we can see that each computer has its own clock speed, motherboard, and CPU processor. The variation in these specifications means that performance can vary widely between different systems. Higher clock speeds can lead to faster processing times, while more advanced motherboards can support additional features and expansion options. Similarly, the type and generation of a CPU processor can greatly influence the efficiency and capability of a computer, affecting everything from multitasking to running high-demand applications. Understanding these differences is essential for choosing the right computer for specific needs, whether for gaming, professional work, or general use.

5.2 Reflection

From this Project part 1, we get to do the benchmarking for our computers. We get to know our laptop specifications after running multiple tests using multiple benchmarking tools such as CPU-Z, Geekbench, Window Task Manager, MiniTool Partition Wizard Free 12.8 and also from assembly codes. We can analyze our laptop specification in terms of clock cycle, motherboard and many other things. We also can compare the performance of our laptop to know which one is better in terms of computer performance compared to the other two.

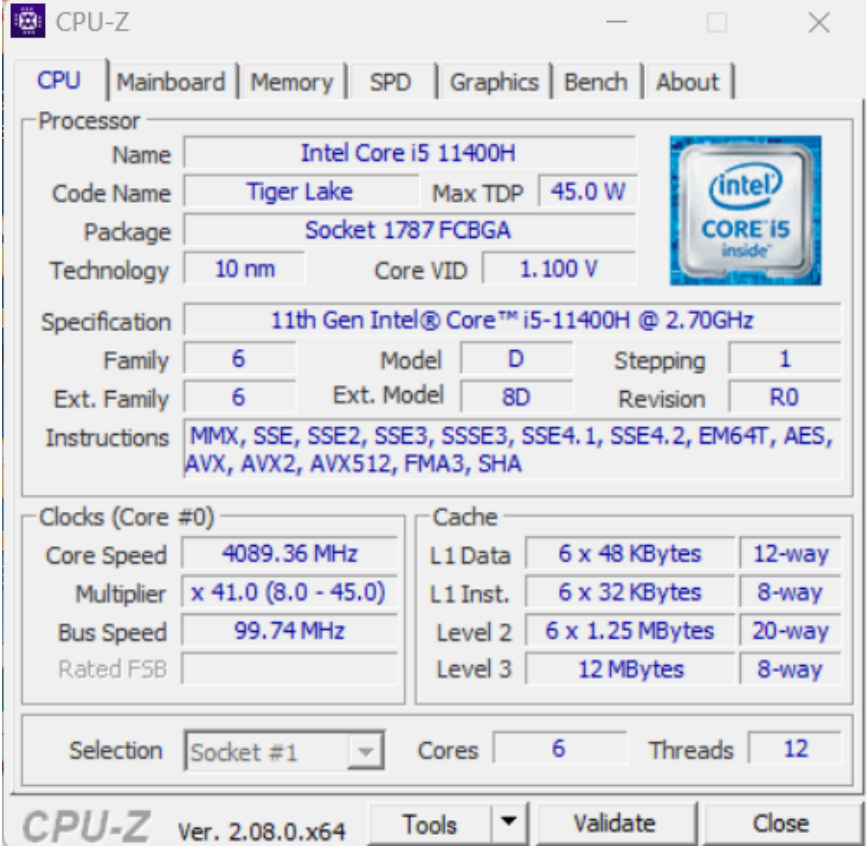
5.3 References

1. Benchmarks as Fast As Possible.
<https://youtu.be/cHtNJngtS1I>
2. Best CPU Benchmark Software For Windows.
<https://youtu.be/fgWGtWsDXmw>
3. First 5 Things I Do When Benchmarking.
<https://youtu.be/19eUZ0GwTxI>
4. What's the Difference Between a CPU and a GPU?
<https://blogs.nvidia.com/blog/2009/12/16/whats-the-difference-between-a-cpu-and-a-gpu/>
5. What is cache memory
<https://www.lenovo.com/my/en/glossary/what-is-cache-memory/?orgRef=https%253A%252F%252Fwww.google.com%252F>
6. CPU benchmark and hierarchy
<https://www.tomshardware.com/reviews/cpu-hierarchy.4312.html>
7. *How to read and understand CPU benchmarks - Intel*. (n.d.). Intel.
<https://www.intel.com/content/www/us/en/gaming/resources/read-cpu-benchmarks.html>
8. Segner, B. (2024, March 18). Using CPU performance benchmarking to determine application sizing. *Medium*.
<https://brentsegner.medium.com/using-cpu-performance-benchmarking-approaches-to-determine-application-sizing-f31eaa2e8da1>
9. An Introduction to CPU Performance Benchmarks and How This Applies to the Home Market
<https://documentation-service.arm.com/static/6194d6b0f45f0b1fbf3a8977?token=>
- 10.

5.4 Appendices

CPU Z

ASUS TUF F15



The screenshot displays the CPU-Z application window, which provides detailed information about the system's processor. The 'CPU' tab is selected, showing the following data:

Processor			
Name	Intel Core i5 11400H		
Code Name	Tiger Lake	Max TDP	45.0 W
Package	Socket 1787 FCBGA		
Technology	10 nm	Core VID	1.100 V
Specification	11th Gen Intel® Core™ i5-11400H @ 2.70GHz		
Family	6	Model	D
Ext. Family	6	Ext. Model	8D
Instructions	MMX, SSE, SSE2, SSE3, SSSE3, SSE4.1, SSE4.2, EM64T, AES, AVX, AVX2, AVX512, FMA3, SHA		

Clocks (Core #0)		Cache	
Core Speed	4089.36 MHz	L1 Data	6 x 48 KBytes
Multiplier	x 41.0 (8.0 - 45.0)	L1 Inst.	6 x 32 KBytes
Bus Speed	99.74 MHz	Level 2	6 x 1.25 MBytes
Rated FSB		Level 3	12 MBytes

Selection	Cores	Threads
Socket #1	6	12

At the bottom of the window, the version 'Ver. 2.08.0.x64' is shown, along with buttons for 'Tools', 'Validate', and 'Close'.

CPU-Z

CPU

Mainboard

Memory

SPD

Graphics

Bench

About

Motherboard

Manufacturer

ASUSTeK COMPUTER INC.

Model

FX506HF

1.0

Bus Specs.

PCI-Express 4.0 (16.0 GT/s)

Chipset

Intel

Tiger Lake-H

Rev.

05

Southbridge

Intel

HM570

Rev.

11

LPCIO

BIOS

Brand

American Megatrends International LLC.

Version

FX506HF.311

Date

04/22/2024

Graphic Interface

Bus

PCI-Express

Current Link Width

Max. Supported

Side Band Addressing

Max. Supported

CPU-Z

Ver. 2.08.0.x64

Tools

Validate

Close

CPU-Z

CPU

Mainboard

Memory

SPD

Graphics

Bench

About

General

Type

DDR4

Channel #

Dual

Size

16 GBytes

DC Mode

Uncore Frequency

3213.5 MHz

Timings

DRAM Frequency

FSB:DRAM

3:48

CAS# Latency (CL)

22.0 clocks

RAS# to CAS# Delay (tRCD)

22 clocks

RAS# Precharge (tRP)

22 clocks

Cycle Time (tRAS)

52 clocks

Row Refresh Cycle Time (tRFC)

560 clocks

Command Rate (CR)

1T

DRAM Idle Timer

Total CAS# (tRDRAM)

Row To Column (tRCD)

CPU-Z

Ver. 2.08.0.x64

Tools

Validate

Close

LENOVO LEGION Y7000

CPU-Z

CPU Mainboard Memory SPD Graphics Bench About

Processor

Name	Intel Core i5 9300H				
Code Name	Coffee Lake	Max TDP	45.0 W		
Package	Socket 1440 FCBGA				
Technology	14 nm	Core VID	0.808 V		
Specification	Intel® Core™ i5-9300HF CPU @ 2.40GHz				
Family	6	Model	E	Stepping	D
Ext. Family	6	Ext. Model	9E	Revision	R0
Instructions	MMX, SSE, SSE2, SSE3, SSSE3, SSE4.1, SSE4.2, EM64T, AES, AVX, AVX2, FMA3				

Clocks (Core #0)

Core Speed	2395.02 MHz
Multiplier	x 24.0 (8.0 - 41.0)
Bus Speed	99.79 MHz
Rated FSB	

Cache

L1 Data	4 x 32 KBytes	8-way
L1 Inst.	4 x 32 KBytes	8-way
Level 2	4 x 256 KBytes	4-way
Level 3	8 MBytes	16-way

Selection Socket #1 Cores 4 Threads 8

CPU-Z Ver. 2.09.0.x64 Tools Validate Close

CPU-Z

CPU Mainboard Memory SPD Graphics Bench About

Motherboard

Manufacturer	LENOVO		
Model	LNVNB161216	SDK0J40700 WIN	
Bus Specs.	PCI-Express 3.0 (8.0 GT/s)		
Chipset	Intel	Coffee Lake	Rev. 0D
Southbridge	Intel	HM370	Rev. 10
LPCIO			

BIOS

Brand	LENOVO
Version	BHCN42WW
Date	05/21/2021

Graphic Interface

Bus	PCI-Express 3.0		
Current Link Width	x16	Max. Supported	x16
Current Link Speed	8.0 GT/s	Max. Supported	8.0 GT/s

CPU-Z Ver. 2.09.0.x64 Tools Validate Close

CPU-Z

CPU

Mainboard

Memory

SPD

Graphics

Bench

About

General

Type

DDR4

Channel #

Single

Size

8 GBytes

DC Mode

Uncore Frequency

2893.9 MHz

Timings

DRAM Frequency

1315.2 MHz

FSB:DRAM

3:40

CAS# Latency (CL)

19.0 clocks

RAS# to CAS# Delay (tRCD)

19 clocks

RAS# Precharge (tRP)

19 clocks

Cycle Time (tRAS)

43 clocks

Row Refresh Cycle Time (tRFC)

467 clocks

Command Rate (CR)

2T

DRAM Idle Timer

Total CAS# (tRDRAM)

Row To Column (tRCD)

CPU-Z

Ver. 2.09.0.x64

Tools

Validate

Close

HP PAVILION 15

CPU-Z

CPU | Mainboard | Memory | SPD | Graphics | Bench | About

Processor

Name	Intel Core i7 1065G7		
Code Name	Ice Lake	Max TDP	15.0 W
Package	Socket 1526 FCBGA		
Technology	10 nm	Core VID	0.712 V



Specification: Intel® Core™ i7-1065G7 CPU @ 1.30GHz

Family	6	Model	E	Stepping	5
Ext. Family	6	Ext. Model	7E	Revision	D1

Instructions: MMX, SSE, SSE2, SSE3, SSSE3, SSE4.1, SSE4.2, EM64T, VT-x, AES, AVX, AVX2, AVX512, FMA3, SHA

Clocks (Core #0)

Core Speed	1297.14 MHz
Multiplier	x 13.0 (4.0 - 39.0)
Bus Speed	99.78 MHz
Rated FSB	

Cache

L1 Data	4 x 48 KBytes	12-way
L1 Inst.	4 x 32 KBytes	8-way
Level 2	4 x 512 KBytes	8-way
Level 3	8 MBytes	16-way

Selection: Socket #1 | Cores: 4 | Threads: 8

CPU-Z Ver. 2.09.0.x64 | Tools | Validate | Close

CPU-Z

CPU | Mainboard | Memory | SPD | Graphics | Bench | About

Motherboard

Manufacturer	HP		
Model	86E2	95.36	
Bus Specs.	PCI-Express 3.0 (8.0 GT/s)		
Chipset	Intel	Ice Lake	Rev. 03
Southbridge	Intel	Ice Lake PCH	Rev. 30
LPCIO			

BIOS

Brand	Insyde		
Version	F.20		
Date	08/16/2023		

Graphic Interface

Bus	PCI-Express		
Current Link Width		Max. Supported	
Side Band Addressing		Max. Supported	

CPU-Z Ver. 2.09.0.x64 | Tools | Validate | Close

CPU-Z

CPU

Mainboard

Memory

SPD

Graphics

Bench

About

General

Type

DDR4

Channel #

Dual

Size

16 GBytes

DC Mode

Uncore Frequency

1097.6 MHz

Timings

DRAM Frequency

1330.4 MHz

FSB:DRAM

1:20

CAS# Latency (CL)

19.0 clocks

RAS# to CAS# Delay (tRCD)

19 clocks

RAS# Precharge (tRP)

19 clocks

Cycle Time (tRAS)

43 clocks

Row Refresh Cycle Time (tRFC)

467 clocks

Command Rate (CR)

1T

DRAM Idle Timer

Total CAS# (tRDRAM)

Row To Column (tRCD)

CPU-Z

Ver. 2.09.0.x64

Tools

▼

Validate

Close

GEEKBENCH

ASUS TUF F15

System Information

Operating System	Microsoft Windows 11 Home Single Language (64-bit)
Model	ASUSTeK COMPUTER INC. ASUS TUF Gaming F15 FX506HF_FX506HF
Motherboard	ASUSTeK COMPUTER INC. FX506HF
Power Plan	Balanced

CPU Information

Name	<u>Intel Core i5-11400H</u>
Topology	1 Processor, 6 Cores, 12 Threads
Identifier	GenuineIntel Family 6 Model 141 Stepping 1
Base Frequency	2.69 GHz
Cluster 1	6 Cores

CPU Information

Maximum Frequency	4491 MHz
Package	Socket 1787 FCBGA
Codename	Tiger Lake
L1 Instruction Cache	32.0 KB x 6
L1 Data Cache	48.0 KB x 6
L2 Cache	1.25 MB x 6
L3 Cache	12.0 MB x 1

Memory Information

Size	16.00 GB
Transfer Rate	3194 MT/s
Type	DDR4 SDRAM
Channels	2

Single-Core Performance

Single-Core Score	2042
File Compression	2036 292.4 MB/sec
Navigation	1785 10.8 routes/sec
HTML5 Browser	2134 43.7 pages/sec
PDF Renderer	2155 49.7 Mpixels/sec
Photo Library	2070 28.1 images/sec
Clang	2068 10.2 Klines/sec
Text Processing	1985 159.0 pages/sec
Asset Compression	1672 51.8 MB/sec
Object Detection	2099 62.8 images/sec
Background Blur	2516 10.4 images/sec
Horizon Detection	2347 73.0 Mpixels/sec
Object Remover	1669 128.3 Mpixels/sec
HDR	1978 58.0 Mpixels/sec
Photo Filter	2326 23.1 images/sec
Ray Tracer	1975 1.91 Mpixels/sec
Structure from Motion	2246 71.1 Kpixels/sec

Multi-Core Performance

Multi-Core Score	8163
File Compression	4652 668.1 MB/sec
Navigation	9802 59.1 routes/sec
HTML5 Browser	8224 168.4 pages/sec
PDF Renderer	10633 245.2 Mpixels/sec
Photo Library	10848 147.2 images/sec
Clang	12402 61.1 Klines/sec
Text Processing	2559 205.0 pages/sec
Asset Compression	11096 343.8 MB/sec
Object Detection	5082 152.1 images/sec
Background Blur	10052 41.6 images/sec
Horizon Detection	10815 336.6 Mpixels/sec
Object Remover	7843 603.0 Mpixels/sec
HDR	8499 249.4 Mpixels/sec
Photo Filter	7037 69.8 images/sec
Ray Tracer	14038 13.6 Mpixels/sec
Structure from Motion	9715 307.6 Kpixels/sec

LENOVO LEGION Y7000

System Information

System Information

Operating System Microsoft Windows 11 Home (64-bit)

Model LENOVO 81T0

Motherboard LENOVO LNVNB161216

Power Plan Balanced

CPU Information

Name [Intel Core i5-9300HF](#)

Topology 1 Processor, 4 Cores, 8 Threads

Identifier GenuineIntel Family 6 Model 158 Stepping 13

Base Frequency 2.40 GHz

Cluster 1 4 Cores

Maximum Frequency 4097 MHz

Package Socket 1440 FCBGA

CPU Information

Codename Coffee Lake

L1 Instruction Cache 32.0 KB x 4

L1 Data Cache 32.0 KB x 4

L2 Cache 256 KB x 4

L3 Cache 8.00 MB x 1

Memory Information

Size 8.00 GB

Transfer Rate 2660 MT/s

Type DDR4 SDRAM

Channels 1

Single-Core Score	1424
File Compression	1332 191.2 MB/sec
Navigation	1670 10.1 routes/sec
HTML5 Browser	1493 30.6 pages/sec
PDF Renderer	1551 35.8 Mpixels/sec
Photo Library	1187 16.1 images/sec
Clang	1554 7.65 Klines/sec
Text Processing	1395 111.7 pages/sec
Asset Compression	1597 49.5 MB/sec
Object Detection	648 19.4 images/sec
Background Blur	1982 8.20 images/sec
Horizon Detection	2114 65.8 Mpixels/sec
Object Remover	1331 102.4 Mpixels/sec
HDR	1341 39.4 Mpixels/sec
Photo Filter	1703 16.9 images/sec
Ray Tracer	1285 1.24 Mpixels/sec
Structure from Motion	1637 51.8 Kpixels/sec

Multi-Core Score	3844
File Compression	1737 249.4 MB/sec
Navigation	5186 31.2 routes/sec
HTML5 Browser	3684 75.4 pages/sec
PDF Renderer	5152 118.8 Mpixels/sec
Photo Library	4032 54.7 images/sec
Clang	6015 29.6 Klines/sec
Text Processing	1773 142.0 pages/sec
Asset Compression	6703 207.7 MB/sec
Object Detection	• 1599 47.8 images/sec
Background Blur	5380 22.3 images/sec
Horizon Detection	5658 176.1 Mpixels/sec
Object Remover	4266 328.0 Mpixels/sec
HDR	3963 116.3 Mpixels/sec
Photo Filter	3016 29.9 images/sec
Ray Tracer	6026 5.83 Mpixels/sec
Structure from Motion	4492 142.2 Kpixels/sec

HP PAVILION 15

System Information	
Operating System:	Microsoft Windows 11 Pro (64-bit)
Model:	HP HP Pavilion Laptop 15-cs3xxx
Motherboard:	HP 86E2
Memory:	16.0 GB DDR4 SDRAM
CPU Information	
Name:	Intel Core i7-1065G7
	1 Processor, 4 Cores, 8 Threads
Codename:	Ice Lake
Package:	Socket 1526 FCBGA
Base Frequency:	1.50 GHz
Maximum Frequency:	3891 MHz

CPU Information	
Name	Intel Core i7-1065G7
Topology	1 Processor, 4 Cores, 8 Threads
Identifier	GenuineIntel Family 6 Model 126 Stepping 5
Base Frequency	1.50 GHz
Cluster 1	4 Cores
Maximum Frequency	3891 MHz
Package	Socket 1526 FCBGA
Codename	Ice Lake
L1 Instruction Cache	32.0 KB x 4
L1 Data Cache	48.0 KB x 4
L2 Cache	512 KB x 4
L3 Cache	8.00 MB x 1

Memory Information

Size	16.00 GB
Transfer Rate	2660 MT/s
Type	DDR4 SDRAM
Channels	2

Single-Core Performance

Single-Core Score	1699	
File Compression	1657 238.0 MB/sec	
Navigation	1696 10.2 routes/sec	
HTML5 Browser	1849 37.8 pages/sec	
PDF Renderer	1763 40.7 Mpixels/sec	
Photo Library	1713 23.3 images/sec	
Clang	1655 8.15 Klines/sec	
Text Processing	1572 125.9 pages/sec	
Asset Compression	1320 40.9 MB/sec	
Object Detection	1686 50.5 images/sec	
Background Blur	2072 8.58 images/sec	
Horizon Detection	2012 62.6 Mpixels/sec	
Object Remover	1376 105.8 Mpixels/sec	
HDR	1657 48.6 Mpixels/sec	
Photo Filter	2007 19.9 images/sec	
Ray Tracer	1630 1.58 Mpixels/sec	
Structure from Motion	1913 60.6 Kpixels/sec	

Multi-Core Performance

Multi-Core Score	3974	
File Compression	2562 368.0 MB/sec	
Navigation	4792 28.9 routes/sec	
HTML5 Browser	4580 93.8 pages/sec	
PDF Renderer	5192 119.7 Mpixels/sec	
Photo Library	4452 60.4 images/sec	
Clang	3901 19.2 Klines/sec	
Text Processing	1859 148.8 pages/sec	
Asset Compression	4530 140.4 MB/sec	
Object Detection	2958 88.5 images/sec	
Background Blur	4391 18.2 images/sec	
Horizon Detection	6103 189.9 Mpixels/sec	
Object Remover	3733 287.0 Mpixels/sec	
HDR	3974 116.6 Mpixels/sec	
Photo Filter	4261 42.3 images/sec	
Ray Tracer	5217 5.05 Mpixels/sec	
Structure from Motion	4682 148.2 Kpixels/sec	

TASK MANAGER

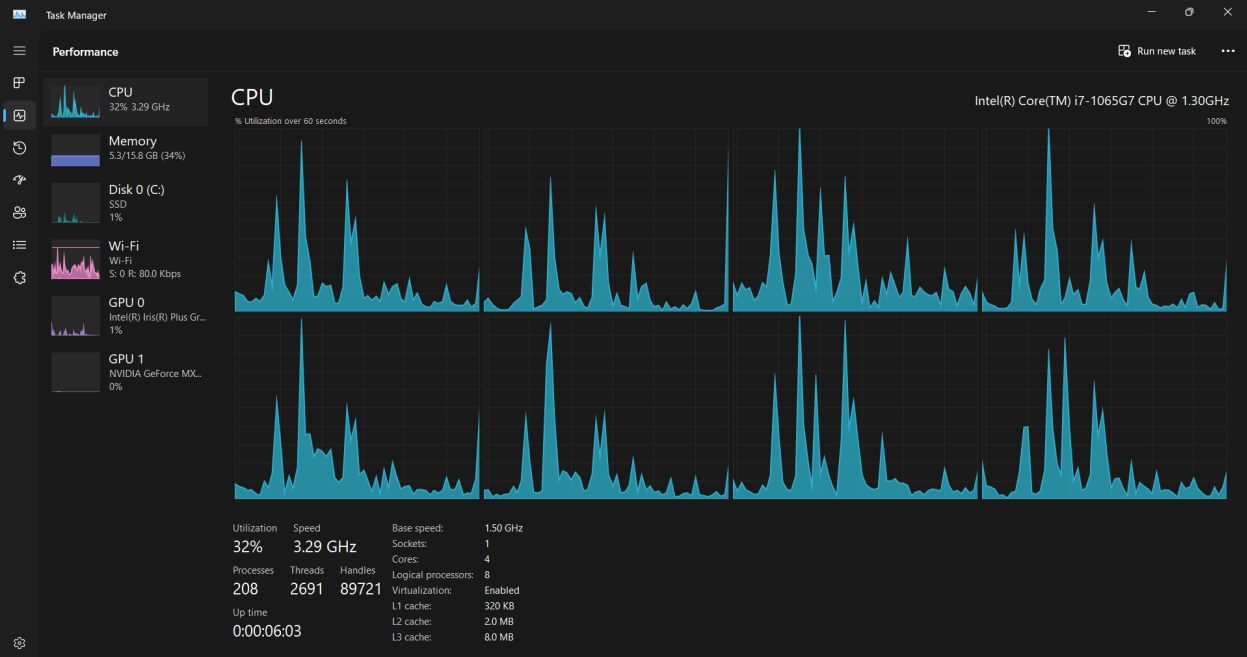
HP PAVILION 15

Task Manager

Type a name, publisher, or PID to search

Processes

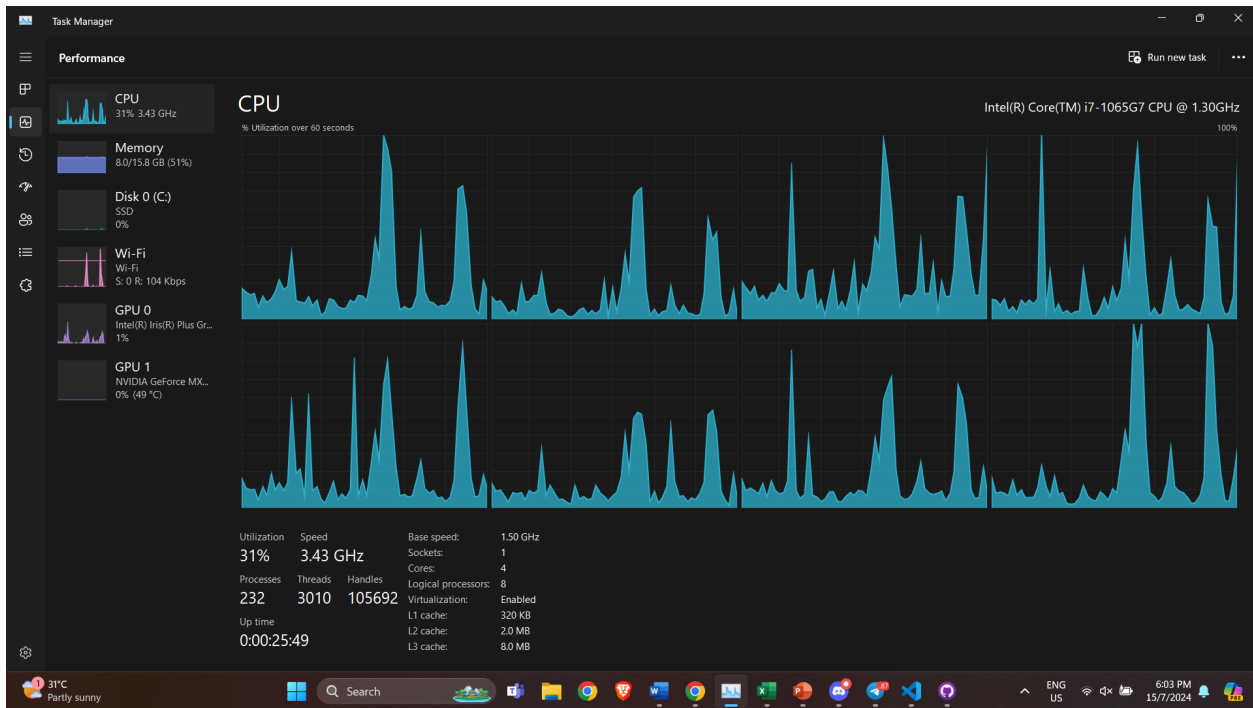
<



Task Manager

Type a name, publisher, or PID to search

Processes



Task Manager

Type a name, publisher, or PID to search

Processes

Performance

App history

Startup apps

Users

Details

Services

Settings

Processes

Name	Status	58% CPU	71% Memory	2% Disk	7% Network
> iw5sp (32 bit)		27.4%	1,244.2 MB	0.1 MB/s	0 Mbps
> Google Chrome (20)		0%	1,221.0 MB	0.1 MB/s	0.1 Mbps
> 07_rectangle.cpp - exercise 1 - ...		0%	469.0 MB	0 MB/s	0 Mbps
> Adobe Acrobat Reader DC (32...		0%	360.8 MB	0 MB/s	0 Mbps
> Friends - Discord (6)		1.2%	304.6 MB	0 MB/s	0 Mbps
> RStudio (7)		0%	287.1 MB	0.1 MB/s	0 Mbps
> Antimalware Service Executable		0%	211.6 MB	0.1 MB/s	0 Mbps
> Telegram Desktop		0%	202.4 MB	0 MB/s	0 Mbps
> Webex (3)		1.8%	189.9 MB	0.1 MB/s	0 Mbps
> Desktop Window Manager		1.1%	180.5 MB	0 MB/s	0 Mbps
> Windows Explorer		0%	169.8 MB	0.1 MB/s	0 Mbps
> C/C++ Extension for Visual Stu...		0%	142.8 MB	0 MB/s	0 Mbps
> GitHub Desktop (4)		0%	118.9 MB	0 MB/s	0 Mbps
> Task Manager		3.5%	73.7 MB	0 MB/s	0 Mbps
> Microsoft PowerPoint		0%	61.8 MB	0 MB/s	0 Mbps
> Adobe RdrCEF (32 bit) (2)		0%	58.1 MB	0.1 MB/s	0 Mbps
> Microsoft Excel		0%	45.9 MB	0 MB/s	0 Mbps
> Microsoft Word		0%	42.1 MB	0 MB/s	0 Mbps
> Microsoft Edge (10)		0%	39.7 MB	0 MB/s	0 Mbps
> wsappx		0%	38.3 MB	0 MB/s	0 Mbps
> Start (2)		0%	26.6 MB	0 MB/s	0 Mbps
> Windows Widgets (8)		0%	21.0 MB	0 MB/s	0 Mbps
> Search (3)		0%	18.1 MB	0 MB/s	0 Mbps
> Windows Audio Device Graph ...		0.6%	16.9 MB	0 MB/s	0 Mbps



ASUS TUF F15

Task Manager

Processes

Performance

App history

Startup apps

Users

Details

Services

Settings

Type a name, publisher, or PID to search

Processes

Run

Name	Status	3% CPU	41% Memory	0% Disk	2% Network
> Google Chrome (10)		0.8%	438.8 MB	0.1 MB/s	0 Mbps
Desktop Window Manager		0.5%	255.5 MB	0 MB/s	0 Mbps
> Antimalware Service Executable		0.4%	223.9 MB	0 MB/s	0 Mbps
Windows Explorer		0%	166.8 MB	0.1 MB/s	0 Mbps
> Microsoft Word		0%	98.2 MB	0 MB/s	0 Mbps
> Task Manager		0.9%	95.2 MB	0 MB/s	0 Mbps
> NVIDIA Share (3)		0%	67.9 MB	0 MB/s	0 Mbps
> NVIDIA Container (3)		0%	55.3 MB	0 MB/s	0 Mbps
> Start (2)		0%	54.5 MB	0 MB/s	0 Mbps
Secure System		0%	47.9 MB	0 MB/s	0 Mbps
> Phone Link (2)		0%	46.3 MB	0 MB/s	0 Mbps
> Microsoft Windows Search Indexer		0%	31.9 MB	0 MB/s	0 Mbps
> Microsoft Office Click-to-Run (SxS)		0%	31.0 MB	0 MB/s	0 Mbps
SnippingTool.exe		0%	29.1 MB	0 MB/s	0 Mbps
DevHome.PI		0%	28.3 MB	0 MB/s	0 Mbps
> NVIDIA Container (2)		0%	25.9 MB	0 MB/s	0 Mbps
> IntelAudioService		0%	21.6 MB	0 MB/s	0 Mbps
Artificial Intelligence (AI) Host for the Microsoft® Windows® Operatin...		0%	20.1 MB	0 MB/s	0 Mbps
> Service Host: UtcSvc		0%	17.6 MB	0 MB/s	0 Mbps
> Service Host: Windows Event Log		0%	16.9 MB	0 MB/s	0 Mbps
> LGHUB Updater		0%	15.7 MB	0 MB/s	0 Mbps
Registry		0%	15.5 MB	0 MB/s	0 Mbps
GHelper		0%	13.4 MB	0.1 MB/s	0 Mbps
> Service Host: State Repository Service		0%	13.1 MB	0 MB/s	0 Mbps
> Microsoft SharePoint		0%	13.1 MB	0 MB/s	0 Mbps
> LocalServiceNoNetworkFirewall (2)		0%	12.1 MB	0 MB/s	0 Mbps

Task Manager

Processes

Performance

App history

Startup apps

Users

Details

Services

Settings

Performance

CPU

14% 3.91 GHz

Memory

6.6/15.7 GB (42%)

Disk 0 (C:)

SSD 0%

Wi-Fi

Wi-Fi 2 S: 8.0 R: 320 Kbps

GPU 0

Intel(R) UHD Graphics 1%

GPU 1

NVIDIA GeForce RTX... 0%

CPU

11th Gen Intel(R) Core(TM) i5-11400H @ 2.70GHz

% Utilization over 60 seconds

Utilization	Speed	Base speed:	2.69 GHz	
14%	3.91 GHz	Sockets:	1	
		Cores:	6	
Processes	Threads	Handles	Logical processors:	12
221	2793	95348	Virtualization:	Enabled
Up time		L1 cache:	480 KB	
0:00:12:36		L2 cache:	7.5 MB	
		L3 cache:	12.0 MB	

Task Manager

Type a name, publisher, or PID to search

Processes

Performance

App history

Startup apps

Users

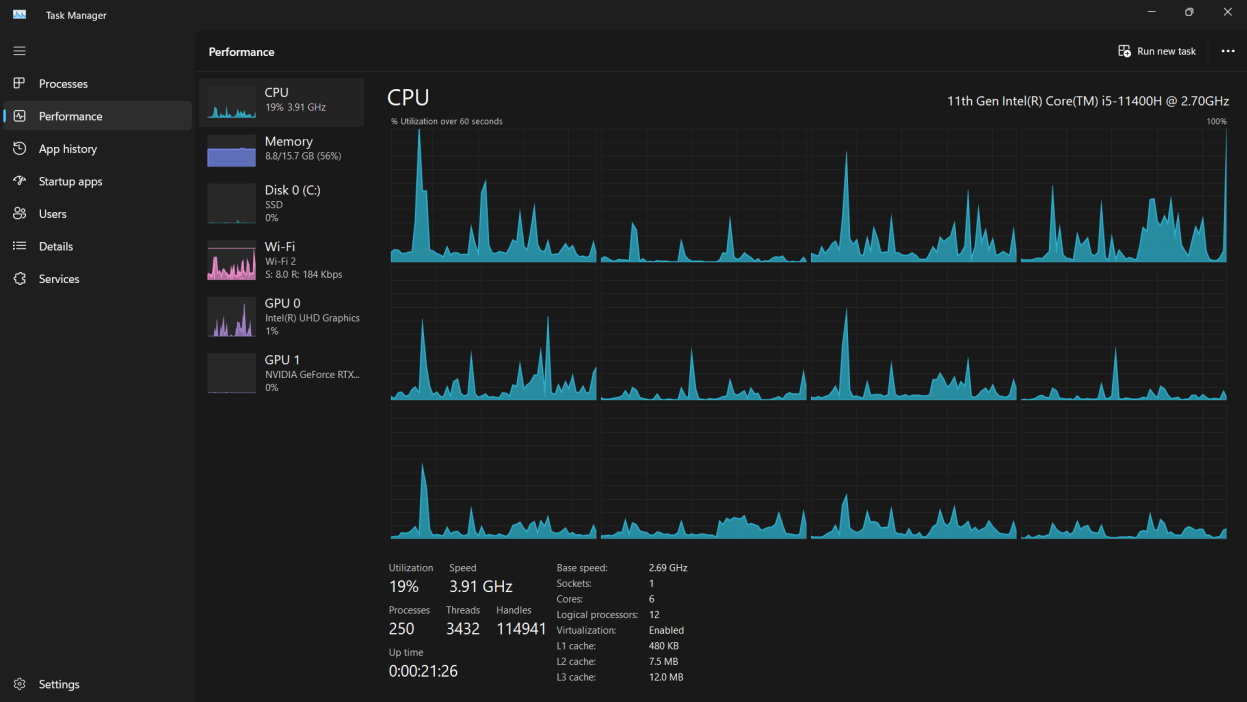
Details

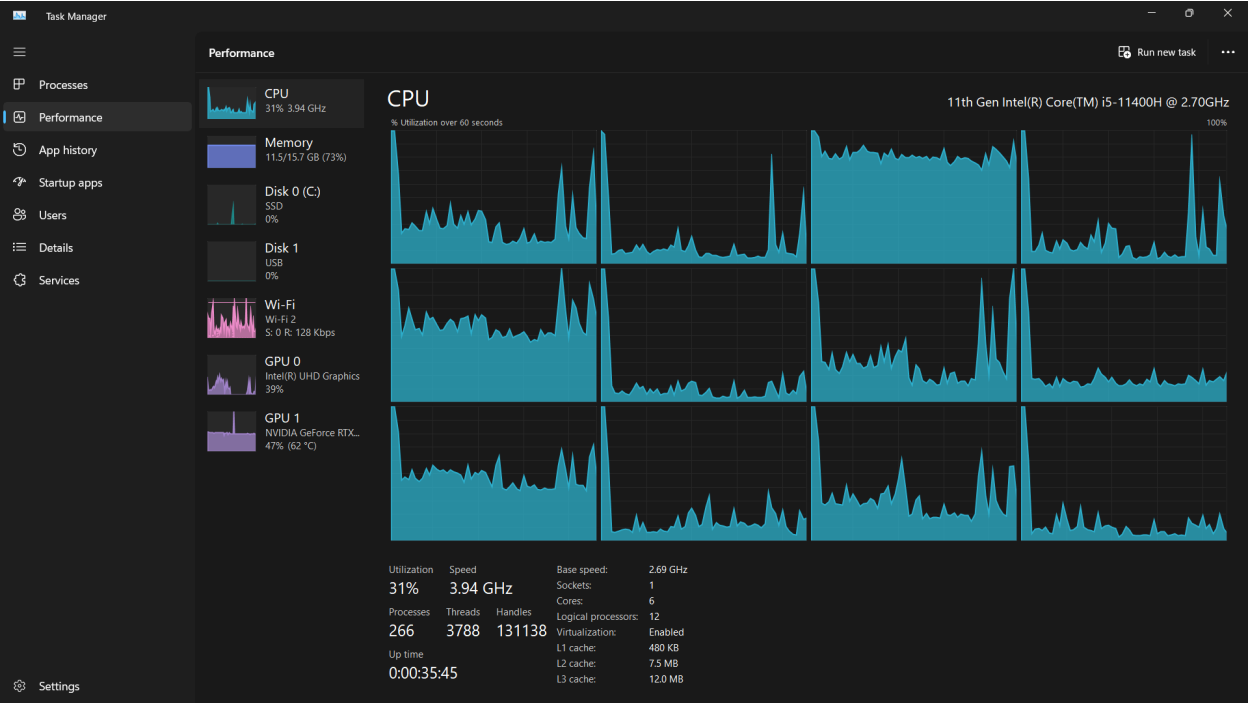
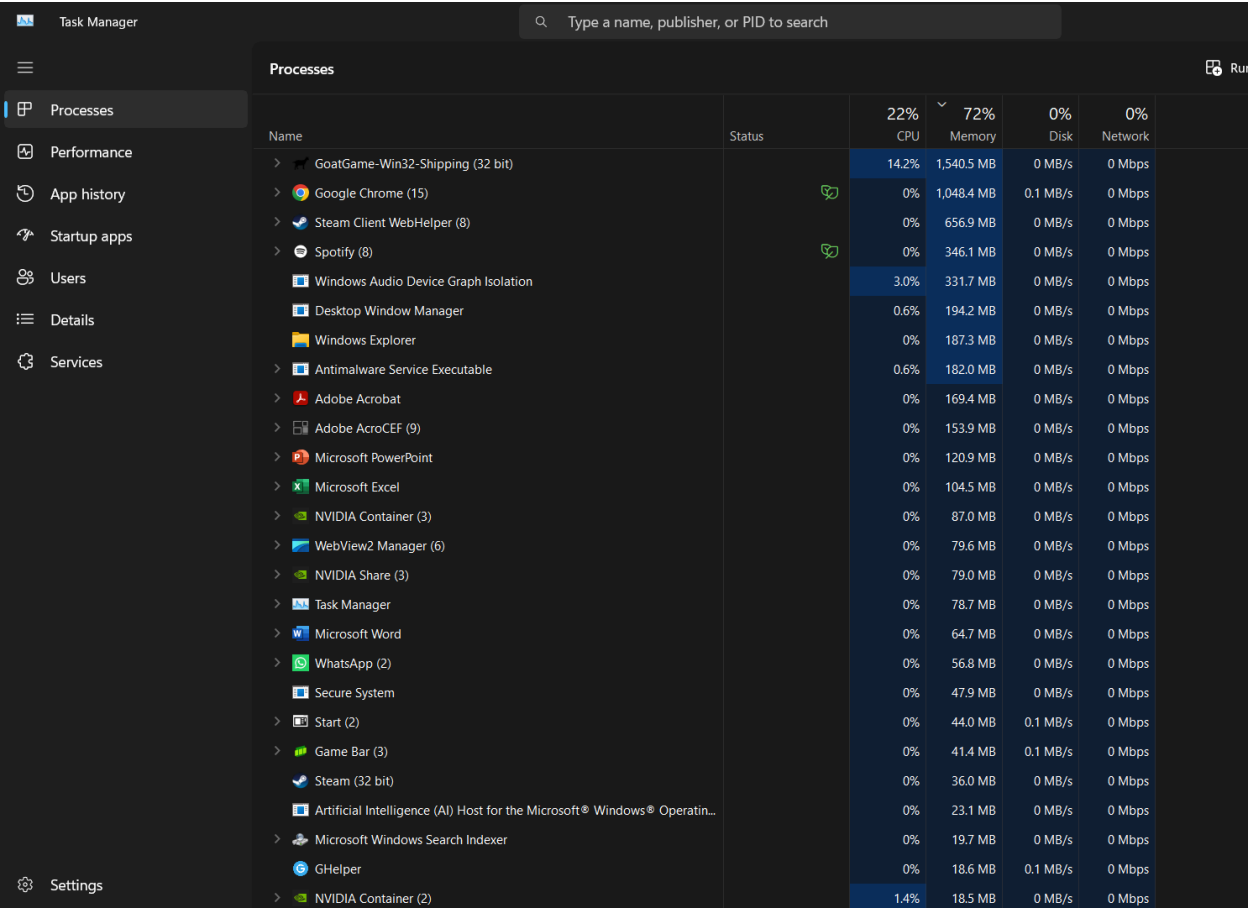
Services

Settings

Processes

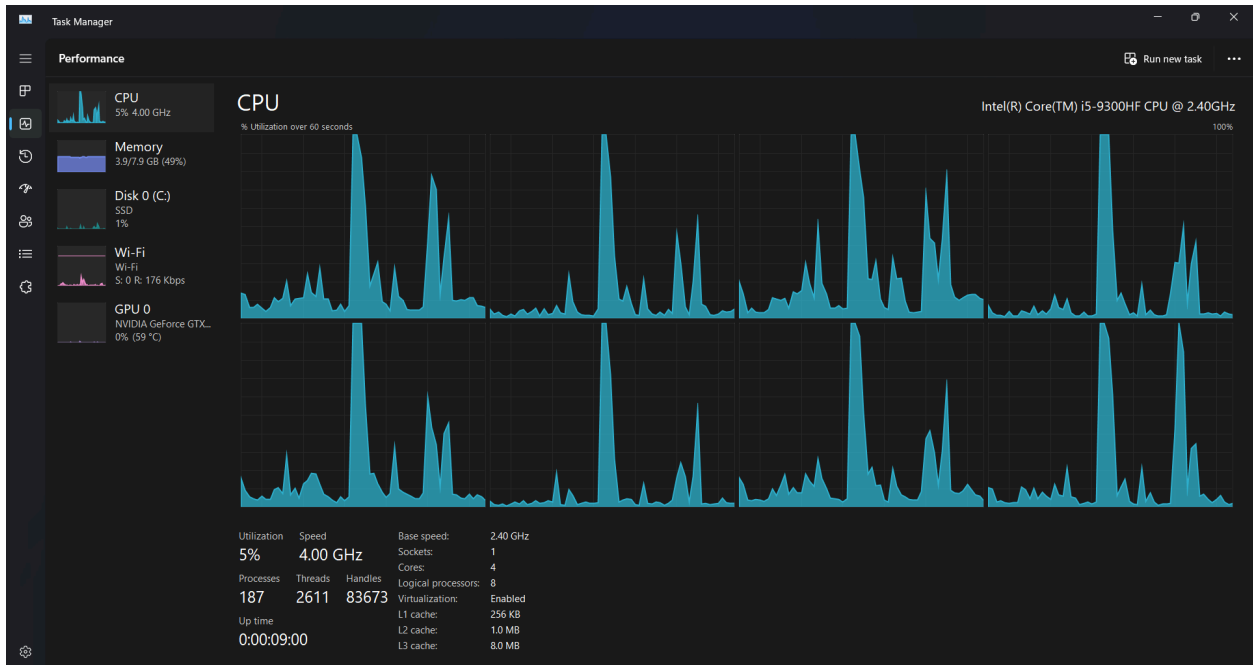
Name	Status	4% CPU	56% Memory	0% Disk	0% Network
> Google Chrome (15)		0%	1,195.8 MB	0.1 MB/s	0.1 Mbps
> Spotify (8)		0%	338.8 MB	0.1 MB/s	0 Mbps
Desktop Window Manager		0%	301.7 MB	0 MB/s	0 Mbps
Windows Explorer		0%	174.1 MB	0 MB/s	0 Mbps
> Adobe Acrobat (2)		0%	169.3 MB	0 MB/s	0 Mbps
> Antimalware Service Executable		0%	156.5 MB	0 MB/s	0 Mbps
> Adobe AcroCEF (9)		0%	156.1 MB	0 MB/s	0 Mbps
> Microsoft PowerPoint		0%	145.6 MB	0 MB/s	0 Mbps
Task Manager		1.7%	108.0 MB	0 MB/s	0 Mbps
> Microsoft Word		0%	102.8 MB	0 MB/s	0 Mbps
> Microsoft Excel		0%	93.9 MB	0 MB/s	0 Mbps
> WebView2 Manager (6)		0%	80.4 MB	0 MB/s	0 Mbps
> NVIDIA Share (3)		0%	68.6 MB	0 MB/s	0 Mbps
> NVIDIA Container (3)		0%	61.0 MB	0.1 MB/s	0 Mbps
> Start (2)		0%	59.1 MB	0 MB/s	0 Mbps
Secure System		0%	47.9 MB	0 MB/s	0 Mbps
> Phone Link (2)		0%	46.2 MB	0 MB/s	0 Mbps
> Microsoft Office Click-to-Run (SxS)		0%	31.3 MB	0 MB/s	0 Mbps
> Microsoft Windows Search Indexer		0%	31.3 MB	0.1 MB/s	0 Mbps
SnippingTool.exe		0%	28.9 MB	0 MB/s	0 Mbps
DevHome.PI		0%	28.3 MB	0 MB/s	0 Mbps
> NVIDIA Container (2)		0%	28.0 MB	0 MB/s	0 Mbps
Artificial Intelligence (AI) Host for the Microsoft® Windows® Operatin...		0%	23.1 MB	0 MB/s	0 Mbps
> IntelAudioService		0%	21.5 MB	0 MB/s	0 Mbps
Artificial Intelligence (AI) Host for the Microsoft® Windows® Operatin...		0%	20.1 MB	0 MB/s	0 Mbps
> Service Host: UtcSvc		0%	19.0 MB	0.1 MB/s	0 Mbps



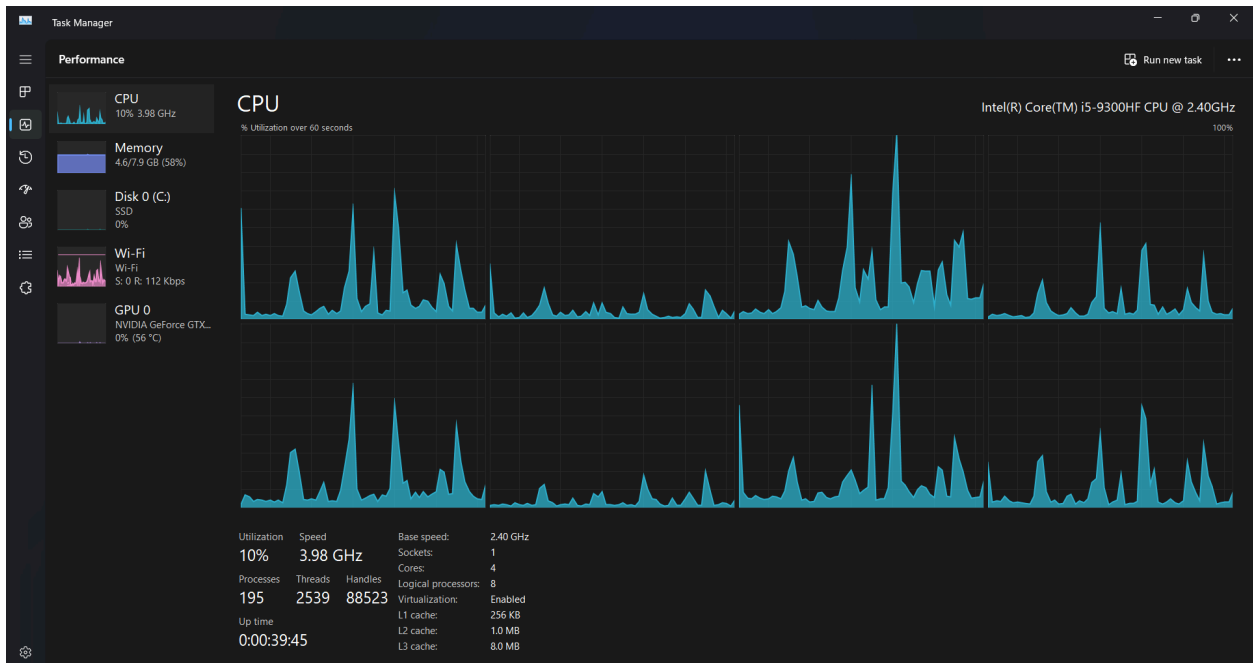


LENOVO LEGION Y7000

Task Manager						
Type a name, publisher, or PID to search						
Processes						
	Name	Status	4% CPU	49% Memory	0% Disk	0% Network
>	Antimalware Service Executable		0%	199.5 MB	0 MB/s	0 Mbps
>	LenovoVantageService	Efficiency ...	0%	109.3 MB	0 MB/s	0 Mbps
	Windows Explorer		0%	65.7 MB	0.1 MB/s	0 Mbps
>	Task Manager		0%	63.6 MB	0.1 MB/s	0 Mbps
>	Microsoft Word		0%	62.6 MB	0 MB/s	0 Mbps
>	NVIDIA Container (3)		0%	53.9 MB	0 MB/s	0 Mbps
>	NVIDIA Share (3)		0%	53.5 MB	0 MB/s	0 Mbps
>	Start (2)		0%	47.1 MB	0 MB/s	0 Mbps
	Desktop Window Manager		0%	40.7 MB	0 MB/s	0 Mbps
	Secure System	Efficiency ...	0%	30.5 MB	0 MB/s	0 Mbps
>	NVIDIA Container (2)	Efficiency ...	0%	27.4 MB	0 MB/s	0 Mbps
>	Search (3)	Suspended	0%	26.5 MB	0 MB/s	0 Mbps
	Lenovo.Modern.ImController.Plugi...	Efficiency ...	0%	26.4 MB	0 MB/s	0 Mbps
>	Microsoft Office Click-to-Run (SxS)	Efficiency ...	0%	26.4 MB	0 MB/s	0 Mbps
	NVIDIA Web Helper Service (32 bit)		0%	22.5 MB	0 MB/s	0 Mbps
	Razer Synapse Service Process (32...		0%	22.3 MB	0.1 MB/s	0 Mbps
>	Lenovo.Modern.ImController (32 b...	Efficiency ...	0%	21.8 MB	0 MB/s	0 Mbps
	Lenovo.Modern.ImController.Plugi...		0%	20.6 MB	0.1 MB/s	0 Mbps
	Artificial Intelligence (AI) Host for ...		0%	20.0 MB	0 MB/s	0 Mbps
>	Service Host: UtcSvc		0%	18.2 MB	0 MB/s	0 Mbps
>	Service Host: Windows Event Log		0%	17.7 MB	0 MB/s	0 Mbps
>	Razer Central Service (32 bit)		0%	15.7 MB	0 MB/s	0 Mbps
	Lenovo.Modern.ImController.Plugi...		0%	14.8 MB	0 MB/s	0 Mbps
>	Microsoft Windows Search Indexer		0%	13.3 MB	0 MB/s	0 Mbps



Task Manager						
Type a name, publisher, or PID to search						
Processes						
	Name	Status	7% CPU	58% Memory	0% Disk	11% Network
>	Google Chrome (12)	Efficiency ...	0%	633.5 MB	0.1 MB/s	0.1 Mbps
>	Antimalware Service Executable		0%	119.1 MB	0 MB/s	0 Mbps
>	Microsoft Excel	Efficiency ...	0%	90.2 MB	0 MB/s	0 Mbps
>	Windows Explorer		0%	68.8 MB	0 MB/s	0 Mbps
>	Task Manager		6.7%	55.6 MB	0 MB/s	0 Mbps
>	NVIDIA Container (3)	Suspended	0%	43.4 MB	0 MB/s	0 Mbps
>	LenovoVantageService	Efficiency ...	0%	42.0 MB	0 MB/s	0 Mbps
>	Microsoft Word		0%	38.7 MB	0 MB/s	0 Mbps
>	Desktop Window Manager		0.3%	35.8 MB	0 MB/s	0 Mbps
>	Secure System	Efficiency ...	0%	30.5 MB	0 MB/s	0 Mbps
>	SnippingTool.exe		0%	28.9 MB	0 MB/s	0 Mbps
>	Start (2)		0%	23.8 MB	0 MB/s	0 Mbps
>	Lenovo.Modern.ImController.Plugi...	Efficiency ...	0%	21.9 MB	0 MB/s	0 Mbps
>	Search (3)	Suspended	0%	19.9 MB	0 MB/s	0 Mbps
>	NVIDIA Share (3)		0%	19.9 MB	0 MB/s	0 Mbps
>	Razer Synapse Service Process (32...		0%	19.5 MB	0.1 MB/s	0 Mbps
>	Service Host: UtcSvc	Efficiency ...	0%	16.8 MB	0 MB/s	0 Mbps
>	NVIDIA Container (2)		0%	15.6 MB	0 MB/s	0 Mbps
>	Lenovo.Modern.ImController (32 ...		0%	14.6 MB	0 MB/s	0 Mbps
>	Microsoft Windows Search Indexer		0%	12.6 MB	0 MB/s	0 Mbps
>	Service Host: Windows Event Log		0%	12.1 MB	0 MB/s	0 Mbps
>	Service Host: Diagnostic Policy Se...		0%	12.0 MB	0 MB/s	0 Mbps
>	Artificial Intelligence (AI) Host for...	Efficiency ...	0%	11.5 MB	0 MB/s	0 Mbps
>	Service Host: State Repository Ser...		0%	10.5 MB	0 MB/s	0 Mbps

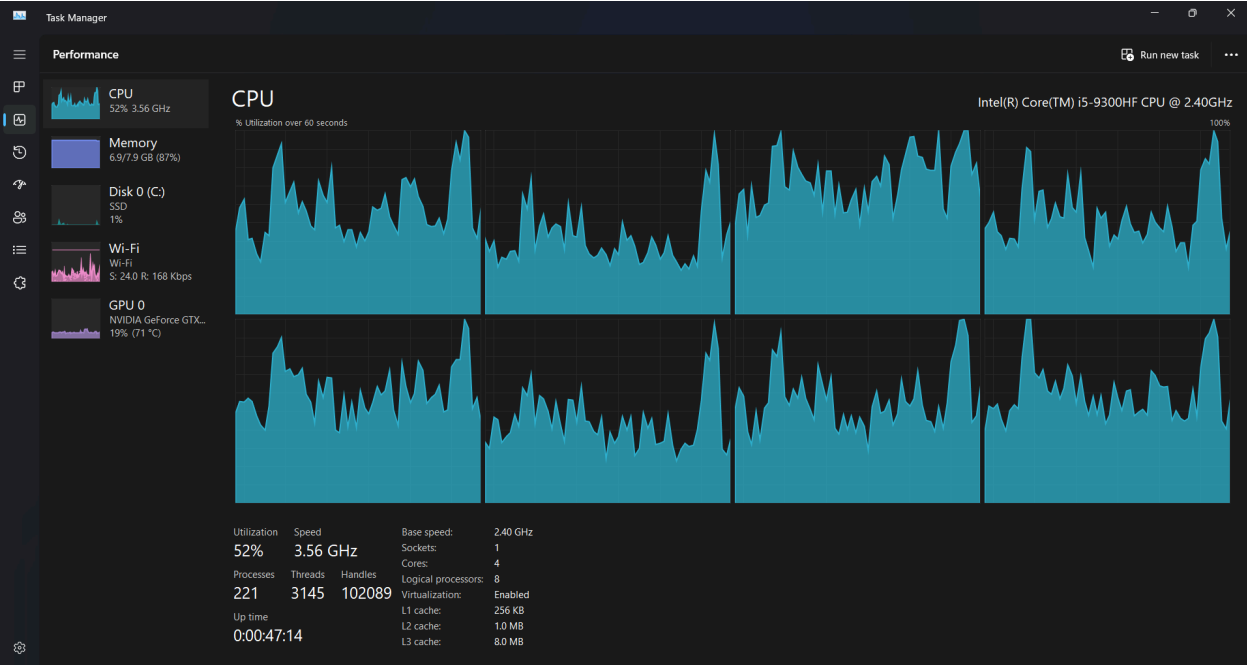


Task Manager

Type a name, publisher, or PID to search

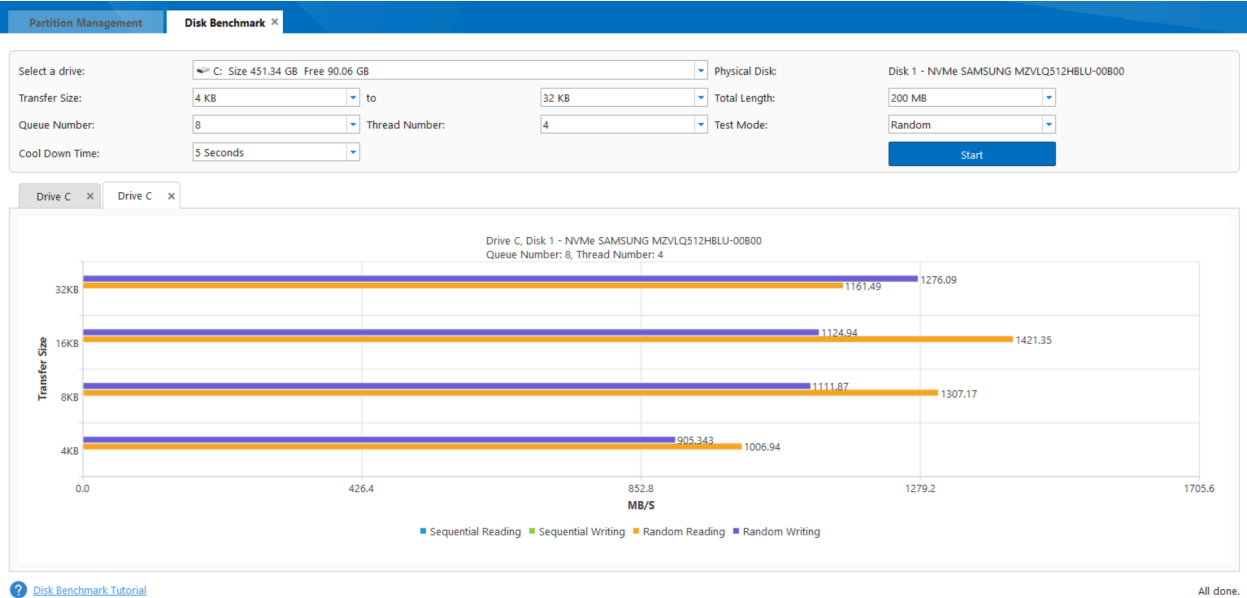
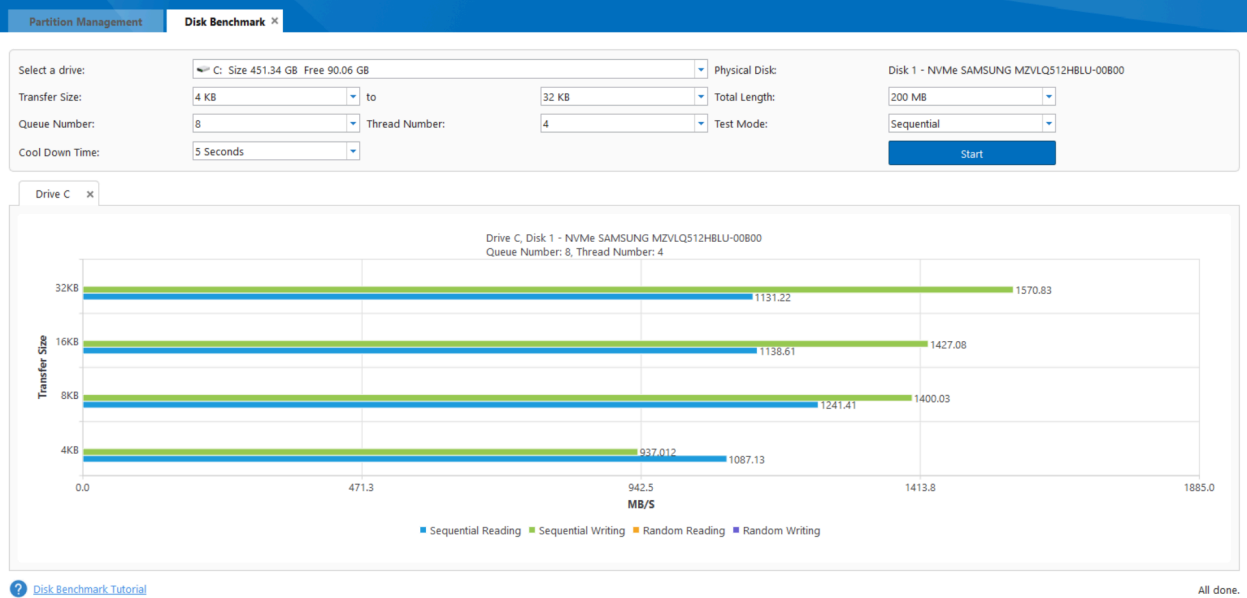
Processes

Name	Status	45% CPU	88% Memory	1% Disk	10% Network
> League of Legends (TM) Client	Efficiency ...	35.9%	720.6 MB	0.1 MB/s	0.1 Mbps
> Google Chrome (12)		0%	577.5 MB	0 MB/s	0.1 Mbps
> League of Legends		0.5%	427.8 MB	0.1 MB/s	0 Mbps
> League of Legends		0%	275.3 MB	0 MB/s	0 Mbps
> Riot Client (32 bit)	Efficiency ...	0.3%	260.0 MB	0.1 MB/s	0.1 Mbps
> Antimalware Service Executable	Efficiency ...	0%	154.4 MB	0.1 MB/s	0 Mbps
> Windows Explorer		0%	117.5 MB	0.1 MB/s	0 Mbps
> League of Legends		1.1%	112.6 MB	0 MB/s	0 Mbps
> Microsoft Excel		0.4%	91.1 MB	0 MB/s	0 Mbps
> NVIDIA Share (3)		0%	61.8 MB	0 MB/s	0 Mbps
> Task Manager		1.2%	58.4 MB	0 MB/s	0 Mbps
> League of Legends		0%	53.2 MB	0 MB/s	0 Mbps
> NVIDIA Container (3)		0%	45.9 MB	0 MB/s	0 Mbps
> Microsoft Word		0%	44.9 MB	0 MB/s	0 Mbps
> LenovoVantageService		0%	44.3 MB	0 MB/s	0 Mbps
> Desktop Window Manager		2.1%	40.1 MB	0 MB/s	0 Mbps
> League of Legends		0%	35.1 MB	0 MB/s	0 Mbps
> Secure System		0%	30.5 MB	0 MB/s	0 Mbps
> Game Bar (3)		0%	29.9 MB	0.1 MB/s	0 Mbps
> Start (2)	0%	28.1 MB	0 MB/s	0 Mbps	
> Lenovo.Modern.ImController.Plugi...	0%	19.8 MB	0 MB/s	0 Mbps	
> Razer Synapse Service Process (32...	0%	19.6 MB	0.1 MB/s	0 Mbps	
> Search (3)	Suspended	0%	19.2 MB	0 MB/s	0 Mbps
> NVIDIA Container (2)		0.3%	19.2 MB	0 MB/s	0 Mbps

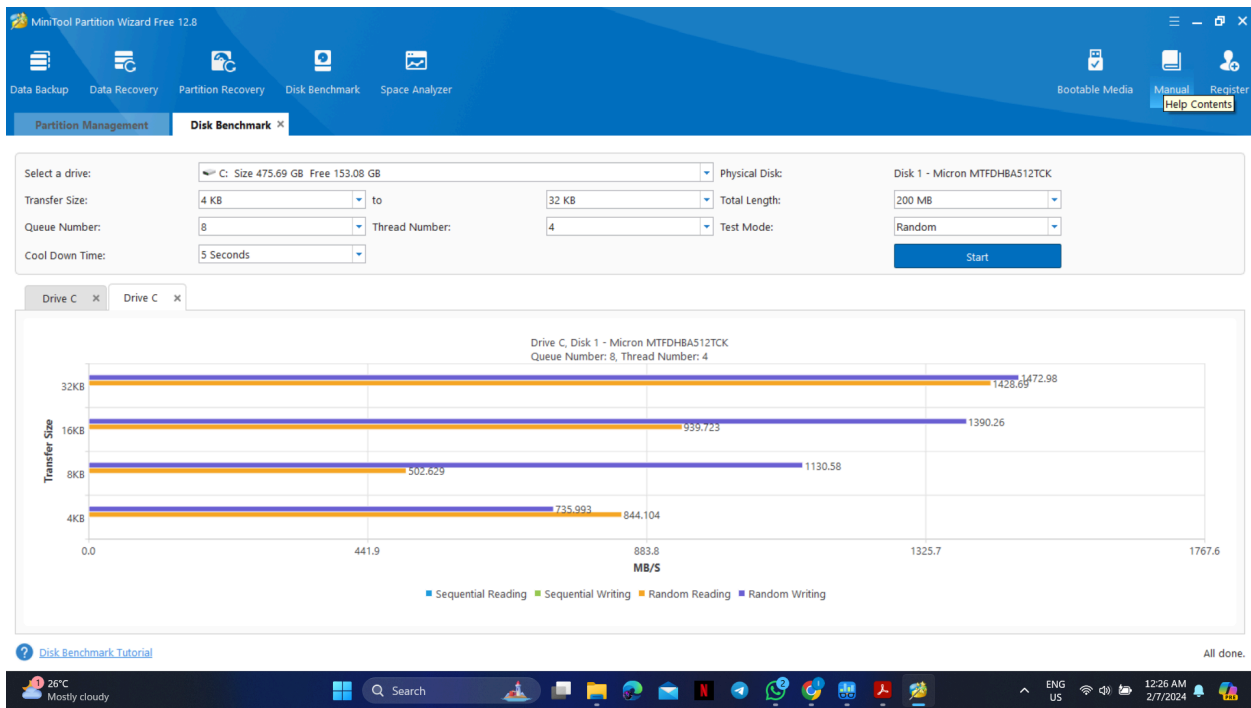
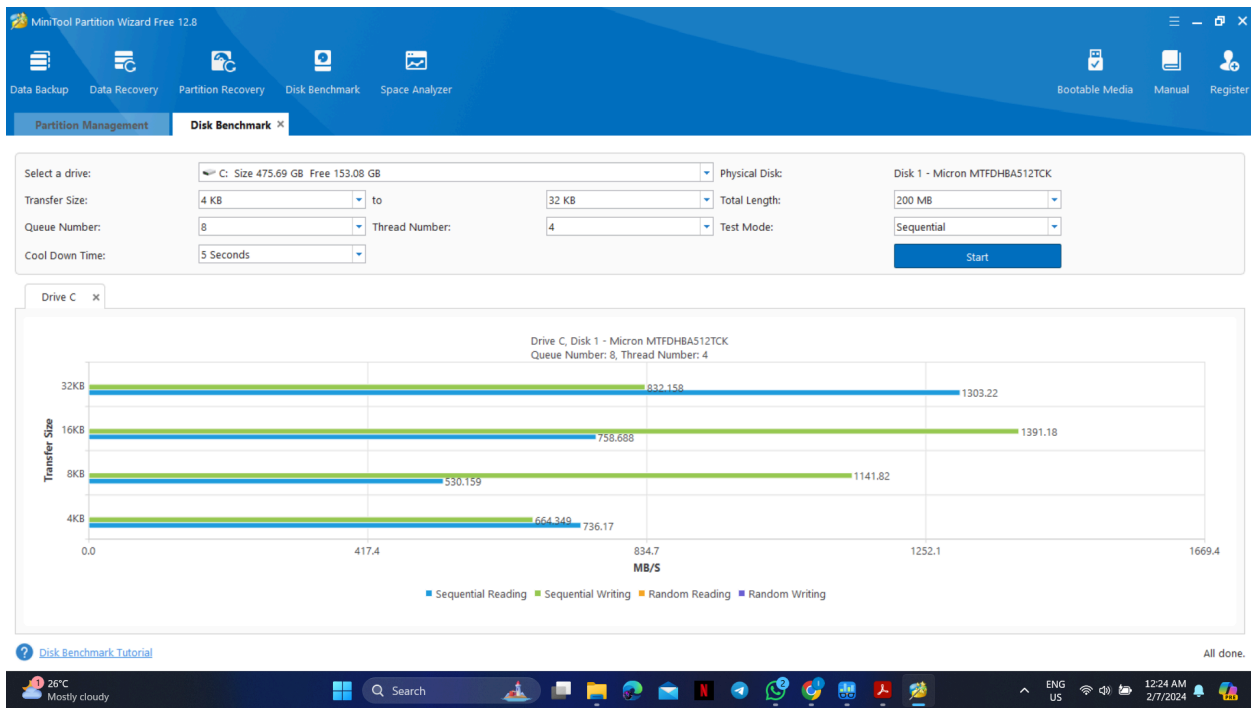


MINI PARTITION WIZARD

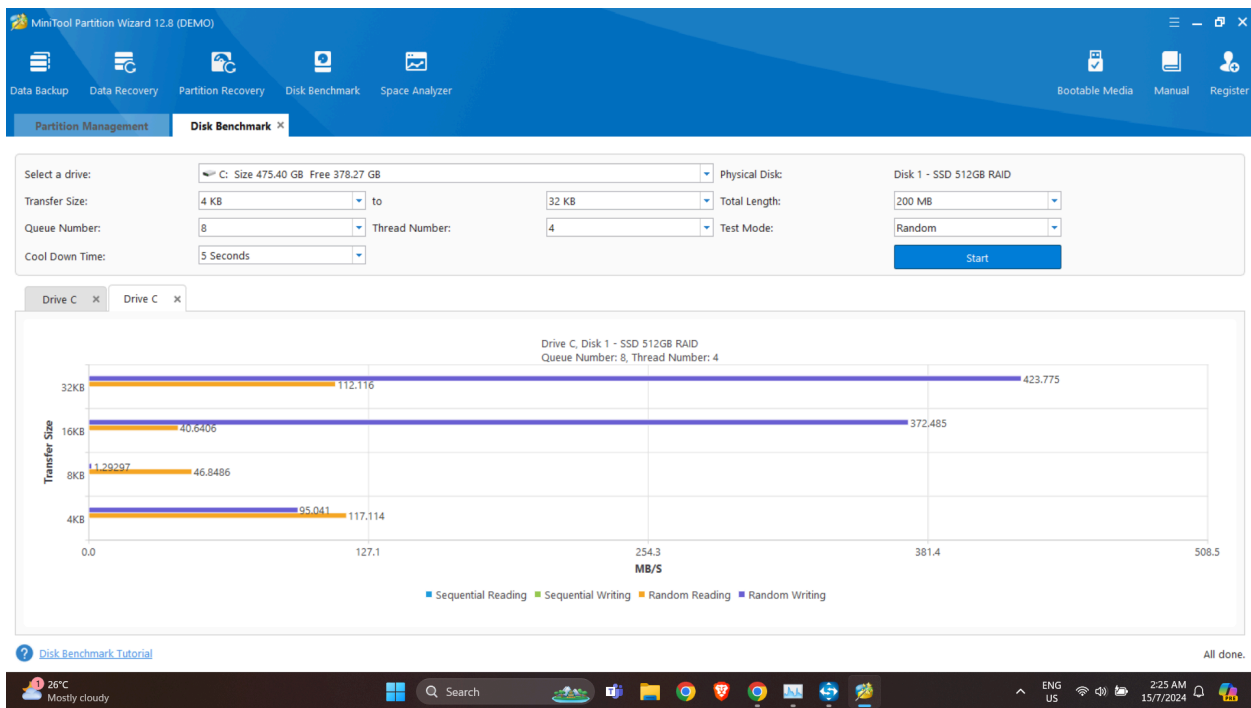
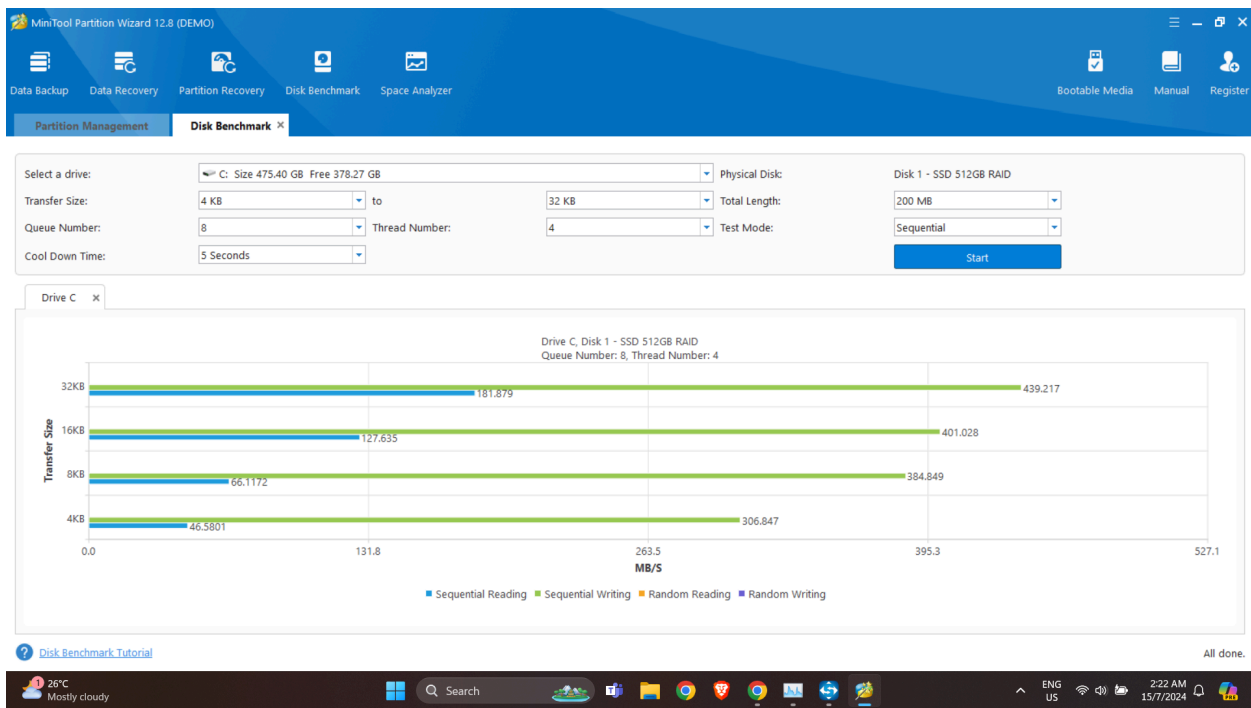
ASUS TUF F15



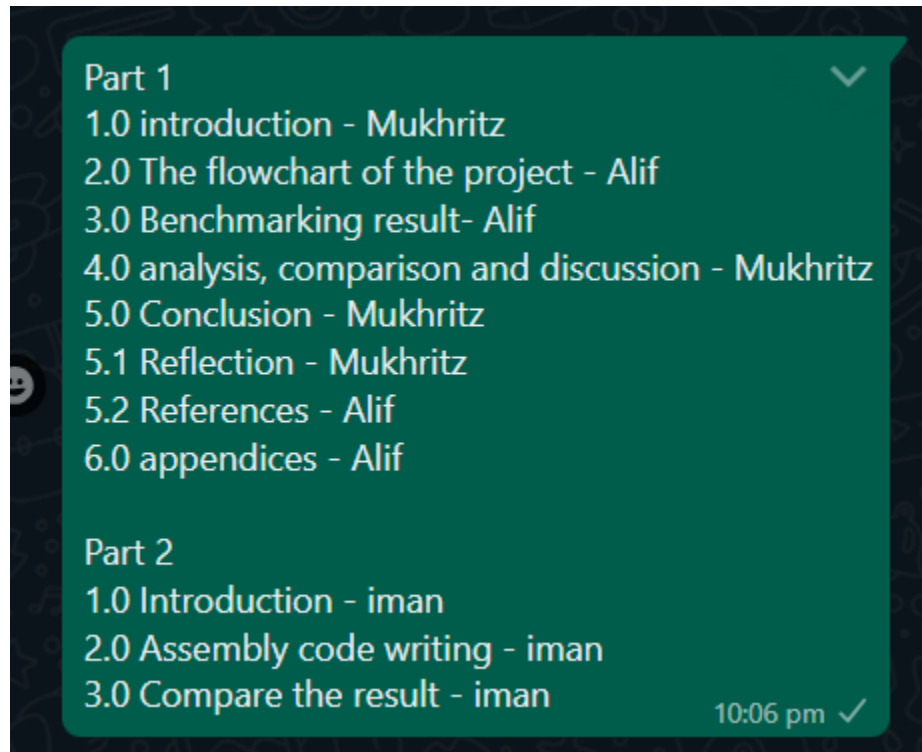
LENOVO LEGION Y7000



HP PAVILION 15



TASK DISTRIBUTION



GROUP PICTURE

